# DevOrSysAdminScripts

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**Abstract**

Some "small" useful scripts for enhancing development or system-administrators tasks

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Current number of commits: 775
Current git SHA1: ba694ba4d8128a0384f8e49a652d4a19a61d4e59
Code lines: 11796 total lines, 10841 not empty lines, 955 empty lines.

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# 1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.libr.php
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
```

```
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpcs_ruleset.xml
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── .gitignore
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├──
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
```

```
│    │ an_see_that_it_works.txt
│    ├── funny.sh
│    ├── spampty_data.txt
│    ├── string_escaping.html
│    ├── string_escaping.xhtml
│    └──
│      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│      that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 83 files

[ 12K Mar 26 22:09]  ./
├── [4.0K Mar 26 22:08]  build_and_checks_dependencies/
│    ├── [4.4K Mar 25 14:39]  build_and_checks.exec.sh.tpl*
│    ├── [1.2K Mar 26 12:30]  build_dependencies_notes.exec.php*
│    ├── [6.9K Mar 26 12:37]  build_dependencies_notes.libr.php
│    ├── [4.2K Mar 25 14:54]  build_md_from_printable_md.exec.sh*
│    ├── [3.3K Mar 25 15:01]  check_shell_scripts_beginnings.libr.sh
│    ├── [5.4K Mar 25 15:35]  check_shell_scripts_indentation.libr.sh
│    ├── [3.6K Mar 23 10:11]  check_URLs.libr.sh
│    ├── [ 31K Mar 26 17:41]  common_build_and_checks.exec.sh*
│    ├── [ 26K Mar 26 17:41]  common_build_and_checks.exec.sh.tpl
│    ├── [ 16K Mar 23 15:17]  common_options.libr.sh
│    ├── [6.1K Mar 22 16:03]  comparisons.libr.sh
│    ├── [ 21K Mar 24 21:57]  create_PDF.exec.sh*
│    ├── [3.7K Mar 22 16:36]  generate_from_template.libr.sh
│    ├── [2.0K Mar 24 20:52]  get_common_text_glob_patterns.libr.sh
│    ├── [3.3K Mar 23 10:18]  grammar_and_spelling_check.libr.sh
│    ├── [4.0K Mar 26 12:19]  licenses_templates/
│    │    ├── [ 14K Mar 26 12:19]  build_licenses_templates.exec.sh*
│    │    ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│    │    ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│    │    └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│    ├── [6.2K Mar 23 15:40]  lines_counts.libr.sh
│    ├── [ 15K Mar 26 11:10]  lines_filters.libr.sh
│    ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│    ├── [2.7K Mar 22 18:33]  overwrite_if_not_equal.libr.sh
│    ├── [2.5K Mar 23 15:29]  python_black_complement.libr.sh
│    ├── [3.0K Mar 23 15:30]  python_isort_complement.libr.sh
```

```
│   ├── [2.1K Mar 23 10:14]  repository_data.libr.sh
│   ├── [4.2K Mar 24 21:29]  shell_checks_complement.libr.sh
│   ├── [2.0K Mar 26 12:37]  split_score.exec.php
│   ├── [2.9K Mar 26 12:37]  split_score.libr.php
│   ├── [ 48K Mar 23 10:16]  strings_functions.libr.sh
│   ├── [4.5K Mar 24 21:31]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar 23 17:15]  update_common_build_and_checks.exec.sh*
│   └── [6.1K Mar 23 17:56]  update_or_check_files_names_listing.exec.sh*
├── [4.6K Mar 26 17:41]  build_and_checks.exec.sh*
├── [4.0K Mar 26 22:09]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [4.7K Mar 25 19:05]  dependencies_data.json
│   ├── [4.0K Mar 25 19:05]  dependencies_notes.txt
│   ├── [5.7K Mar 25 19:01]  DevOrSysAdminScripts.html.tpl
│   ├── [ 34K Mar 26 22:07]  DevOrSysAdminScripts.tex
│   ├── [4.2K Mar 25 19:01]  DevOrSysAdminScripts.tex.tpl
│   ├── [5.5K Mar 26 22:08]  files_names_listing.txt
│   ├── [  23 Mar 24 19:33]  .gitignore
│   ├── [1.4K Mar 25 19:00]  licenses_notes.txt
│   ├── [ 12K Mar 26 22:04]  phpcs_ruleset.xml
│   ├── [2.2K Mar 26 12:12]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.5K Mar 22 15:51]  repository_data.txt
│   ├── [ 786 Mar 22 15:39]  repository_data.txt.tpl
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 166 Mar 24 19:33]  shellcheck.ini
│   └── [3.5K Mar  9 14:46]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [461K Mar 26 22:07]  DevOrSysAdminScripts.html
├── [401K Mar 26 22:07]  DevOrSysAdminScripts.pdf
├── [4.0K Mar 24 21:26]  git_helpers/
│   ├── [3.8K Mar 22 21:06]  git_checks.libr.sh
│   ├── [2.1K Mar 22 21:09]  post-commit*
│   ├── [ 15K Mar 24 21:26]  post_commit_renaming_comment.libr.sh
│   └── [1.8K Mar 22 21:11]  pre-commit*
├── [   8 Mar 26 10:37]  .gitignore
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar 18 17:27]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
```

```
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 01234\
    │── [   0 Mar  4 10:01]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345\
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456\
    │── [   0 Mar  4 09:13]
    │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
    │ an_see_that_it_works.txt
    │── [1.3K Mar  9 15:15]  funny.sh
    │── [  94 Mar  3 19:56]  spampty_data.txt
    │── [ 442 Mar 18 16:50]  string_escaping.html
    │── [ 654 Mar 18 17:27]  string_escaping.xhtml
    └── [   0 Mar  4 09:13]
      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
      that_you_can_see_that_it_works.txt
├── [8.1K Mar 22 21:29]  README.html
├── [4.3K Mar 22 21:29]  README.md
├── [4.3K Mar 22 21:29]  README.md.tpl
├── [7.0K Mar 26 20:09]  string_escaping.libr.php
├── [ 16K Mar 26 20:11]  unicode.libr.php
├── [1.8K Mar 23 18:45]  upgrade_build_and_checks.exec.sh*
└── [2.5K Mar 23 18:45]  wget_sha512.libr.sh

6 directories, 83 files
```

## 2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:

basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;

- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/,
https://savannah.gnu.org/projects/findutils/.

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/,
https://savannah.gnu.org/projects/grep/.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,

https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;) ).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

# 3 Listing of files

## 3.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
```

```
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
fi
readonly LFBFL_verbose

local LFBFL_check_download=""
if [[ "$*" == *--check-download=keep* ]]; then
  LFBFL_check_download="--check-download=keep"
elif [[ "$*" == *--check-download* ]]; then
  LFBFL_check_download="--check-download"
fi
readonly LFBFL_check_download

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
mkdir --parents "${LFBFL_subdir}/licenses_templates"
local LFBFL_variables_directory
LFBFL_variables_directory="${LFBFL_work_directory}/"
LFBFL_variables_directory+="build_and_checks_variables"
readonly LFBFL_variables_directory
mkdir --parents "${LFBFL_variables_directory}/temp"
if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
  printf "temp/\nupgrade_venvs_ts\n"\
    > "${LFBFL_variables_directory}/.gitignore"
fi

local LFBFL_dependencies_URL
LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
readonly LFBFL_dependencies_URL

declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
local LFBFL_script_download_URL
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
LFBFL_script_download_URL+="${LFBFL_common_file_name}"
readonly LFBFL_script_download_URL
declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
@sha512_common_build_and_checks.exec.sh@
wget_sha512 "${LFBFL_file_path}"\
  "${LFBFL_script_download_URL}"\
  "${LFBFL_correct_sha512}"\
  "${LFBFL_verbose}"\
  "${LFBFL_check_download}"
chmod +x "${LFBFL_file_path}"
```

```

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

### 3.2 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php

/**
This file is part of DevOrSysAdminScripts library.

```

```
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@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once 'string_escaping.libr.php';
require_once 'build_dependencies_notes.libr.php';

build_dependencies_notes($argv[1]);
?>
```

### 3.3 build_and_checks_dependencies/build_dependencies_notes.libr.php

```
<?php

/**
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```

```
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2026-03-26T11:59+01:00: This file was renamed from
"build_dependencies_notes.exec.php"
to
"build_dependencies_notes.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

// require_once 'string_escaping.libr.php';



/**
This function looks at the dependencies data found in a work directory,
and generate three files: a txt file, a TeX file (.sub),
and an HTML file (.sub) with the data in a nicer format.

@param string $s_work_directory The directory where the work must be done.

@return void
*/
```

```
function build_dependencies_notes(string $s_work_directory) : void {
  $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
  $s_subdir2 = $s_subdir1.'/temp';
  $s_data_file_path = $s_subdir1.'/dependencies_data.json';
  $s_data = file_get_contents($s_data_file_path);
  $arr_data = json_decode($s_data, true);
  $s_result_content1 = '';
  $s_result_content2 = '';
  $s_result_content3 = '';

  foreach($arr_data as $arr_dependency){
    $s_URL_word = 'URL';
    if(count($arr_dependency['URLs']) > 1){
      $s_URL_word .= 's';
    }

    $s_result_content3 .= "<p>\n";

    $s_dependency1 = $arr_dependency['dependency_name'];
    $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
    $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);

    $s_sentence_end = ' can be found at the following '.$s_URL_word.':';
    $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
    $s_result_content2 .= $s_dependency2.$s_sentence_end."\\newline\n";
    $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";

    for(
      $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
      $i <= $i_max;
      ++$i
    ){
      $mixed_URL = $arr_dependency['URLs'][$i];
      $s_comment = '';
      $s_URL = '';
      if(is_array($mixed_URL)){
        $s_URL = $mixed_URL['URL'];
        $s_comment = ' '.$mixed_URL['comment'];
      }
      else{
        $s_URL = $mixed_URL;
      }

      $s_result_content1 .= $s_URL.$s_comment."\n";

      if(strlen($s_URL) > 60){
```

```
        $s_result_content2 .= (
          "{\\small\n"
          .'  \url{'.$s_URL."}\n"
          .'}'
        );
      }
      else{
        $s_result_content2 .= '\url{'.$s_URL.'}';
      }
      $s_result_content2 .= string_escaping\TeX\escape_text($s_comment);
      if($i === $i_max){
        $s_result_content2 .= ".\n";
      }
      else{
        $s_result_content2 .= ",\\newline\n";
      }

      $s_result_content3 .= (
        "<a\n"
        .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
        ."><span style=\"font-family:monospace\"\n"
        .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
        .'></a>'
      );
      $s_result_content3 .= string_escaping\HTML\escape_text($s_comment);
      if($i === $i_max){
        $s_result_content3 .= ".\n";
      }
      else{
        $s_result_content3 .= ",<br>\n";
      }
    }//end for urls

    $s_result_content1 .= "\n";
    $s_result_content2 .= "\n";
    $s_result_content3 .= "</p>\n\n";
  }//end foreach($arr_data as $arr_dependency)

  // Add a "footer".
  $s_closing_sentence = (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content1 .= $s_closing_sentence;
  $s_result_content2 .= $s_closing_sentence;
  $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";
```

```
// We overwrite results if needed.
$s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
$s_existing_result_content1 = '';
if(file_exists($s_result_path1)){
  $s_existing_result_content1 = file_get_contents($s_result_path1);
}
if($s_result_content1 !== $s_existing_result_content1){
  file_put_contents($s_result_path1, $s_result_content1);
}
/*
Example of generated content of dependencies_notes.txt:
------------------------------------------------------------------------
more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?
------------------------------------------------------------------------
*/
$s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
$s_existing_result_content2 = '';
if(file_exists($s_result_path2)){
  $s_existing_result_content2 = file_get_contents($s_result_path2);
}
if($s_result_content2 !== $s_existing_result_content2){
  file_put_contents($s_result_path2, $s_result_content2);
}
/*
Example of generated content of dependencies_notes.tex.sub:
------------------------------------------------------------------------
more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},\newline
\url{https://github.com/util-linux/util-linux} official?.
------------------------------------------------------------------------
*/
$s_result_path3 = $s_subdir2.'/dependencies_notes.html.sub';
$s_existing_result_content3 = '';
if(file_exists($s_result_path3)){
  $s_existing_result_content3 = file_get_contents($s_result_path3);
}
if($s_result_content3 !== $s_existing_result_content3){
  file_put_contents($s_result_path3, $s_result_content3);
}
```

```
  /*
  Example of generated content of dependencies_notes.html.sub:
  <p>
  more of util-linux can be found at the following URLs:<br>
  <a
  href="https://www.kernel.org/pub/linux/utils/util-linux/"
  ><span style="font-family:monospace"
  >https://www.kernel.org/pub/linux/utils/util-linux/</span
  ></a>,<br>
  <a
  href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
  ><span style="font-family:monospace"
  >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
  ></a>,<br>
  <a
  href="https://github.com/util-linux/util-linux"
  ><span style="font-family:monospace"
  >https://github.com/util-linux/util-linux</span
  ></a> official?.
  </p>
  */
}//end build_dependencies_notes()
?>
```

### 3.4 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
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```

```
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "Searching md file: %s.\n" "${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
```

```
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 1
  fi

  pandoc --from=markdown --to=html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    --output="${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
```

```
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

### 3.5 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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#
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
```

```
  # $1=file_path
  #
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head --lines=1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head --lines=1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head --lines=1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
```

```
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head --lines=1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

### 3.6 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
```

```
32    # $1=file_path
33    declare -i LFBFL_i_file_with_error=0
34    local LFBFL_some_line
35    local LFBFL_previous_line="workeduntilsomeasshole"
36    # created a command named "workeduntilsomeasshole"
37    # and called it without neither indentation, nor parameters
38    # in a .sh script XD.
39    # Manual justification of the lines L-4 and L-3 was a serendipity.
40    # I just tried a joke and was lucky ;) :).
41    local LFBFL_line_end
42    local LFBFL_previous_line_end
43    declare -i LFBFL_i_line_length
44    declare -i LFBFL_i_line_end_length
45    declare -i LFBFL_i_line_start_length
46    local LFBFL_spacing
47    declare -i LFBFL_i_offset
48    local LFBFL_substring1
49    local LFBFL_substring2
50    declare -r LFBFL_pattern='^ *-e '
51
52    # This grep would have been enough without some sed commands.
53    declare -r LFBFL_s_lines=$(
54      grep --before-context=1 --extended-regexp --line-number\
55        --regexp='^(  )* ([^ ]|$)' --with-filename "$1"
56    )
57    if [[ -z "${LFBFL_s_lines}" ]]; then
58      return
59    fi
60
61    declare -a LFBFL_arr_lines
62    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
63    readonly LFBFL_arr_lines
64
65    for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
66      if [[ "${LFBFL_some_line}" == "--" ]]; then
67        # When switching between code fragments, we reinit.
68        LFBFL_previous_line="workeduntilsomeasshole"
69        continue
70      fi
71      # Note how it's funny that this loop always work because we check
72      # shell scripts beginning in another script and thus we always
73      # have a line before.
74      if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]]; then
75        LFBFL_previous_line="${LFBFL_some_line}"
76        continue
77      fi
```

```
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_i_line_length=${#LFBFL_some_line}
    LFBFL_i_line_end_length=${#LFBFL_line_end}
    LFBFL_i_line_start_length=$((
      LFBFL_i_line_length - LFBFL_i_line_end_length
    ))
    LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_i_offset += 1));;
      :) ((LFBFL_i_offset += 1));;
      [0-9]) ((LFBFL_i_offset += 2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_i_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
```

```
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

### 3.7 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
```

```
# or (at your option) any later version.
#
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#
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )
```

```

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"\
      | grep -v 'http://www.w3.org/1999/xhtml'\
      | grep -v 'http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'
    # Second grep after xargs grep just above will remove false positives
    # from substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
      fi
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_path}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            "${LFBFL_file_path}"
        fi
      done
    done
```

```
  done
}
```

### 3.8 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
```

```
LFBFL_work_directory_option+="${LFBFL_work_directory}"
declare -r LFBFL_dependencies_URL="$2"
local LFBFL_verbose=""
declare -i LFBFL_i_verbose=0
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
  LFBFL_i_verbose=1
fi
readonly LFBFL_verbose
readonly LFBFL_i_verbose
local LFBFL_check_download=""
if [[ "$*" == *--check-download=keep* ]]; then
  LFBFL_check_download="--check-download=keep"
elif [[ "$*" == *--check-download* ]]; then
  LFBFL_check_download="--check-download"
fi
readonly LFBFL_check_download

declare -a LFBFL_some_common_options=(
  "${LFBFL_verbose}"
  # "$--work-directory={LFBFL_work_directory}" doesn't work
  "${LFBFL_work_directory_option}"
)

source ./wget_sha512.libr.sh

local LFBFL_subdir="build_and_checks_dependencies"
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_subdir3="build_and_checks_variables"
local LFBFL_file_name
local LFBFL_script_download_URL
local LFBFL_file_path
local LFBFL_correct_sha512

wrapped_wget_sha512(){
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

LFBFL_file_name="build_md_from_printable_md.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
LFBFL_correct_sha512='ceae57a25ad40c149a7a020ddbf5e85941f4212e2ffd0'
LFBFL_correct_sha512+='1f2b00b14c895aeb628cca53565b0afacb8f1e24dbbc'
LFBFL_correct_sha512+='39b270cdc0c61e4cfe42bafd0d9b6bec4215405'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='624368d161351d7a261fbe3165d51907f46394c1b0e79'
LFBFL_correct_sha512+='b980ec11173c324b2c5531cf2a2b60b2dc62edcaa9c8'
LFBFL_correct_sha512+='71dca55065f8802df4275420836252786f9606b'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='6a57c0f30a0f6f15e9375384bb0ff214bdcbc3377c238'
LFBFL_correct_sha512+='e33febdb3a905d96358c7392b7b66c71aad022435527'
LFBFL_correct_sha512+='00bf10f30dfeb0db9fc84d06eba7343c1e06184'
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='86bccfa7181772622db577cbfe0d2be7c4ea8a8c3550e'
LFBFL_correct_sha512+='5edcd21b20628dd5f5539c287a2af9b7a27933a7c812'
LFBFL_correct_sha512+='3a26c057bbd54175b89917336830529eb3831bd'
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2027e3f9cdb847cc71b4b69149486779df666a1b18c30'
LFBFL_correct_sha512+='ab161b7c96ee2c48fed4ef78edb489d566d568fb7b1c'
LFBFL_correct_sha512+='8390a680a5212f5a04c9bd6dd05744374bbc5b3'
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='58f125f10f9c10cd38f04cdeec997bb7dc376ba6f0c09'
LFBFL_correct_sha512+='dfc1d2c0927855d709fbf002c027e67370cc0f8864b7'
LFBFL_correct_sha512+='8b23ab7685fc192413cc657df46475e28b989e2'
wrapped_wget_sha512
```

```
134
135     LFBFL_file_name="comparisons.libr.sh"
136     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
137     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
138     LFBFL_correct_sha512='eb2bfabf11d219a4f7a105c95ea5fc2b1f6ddd668fad7'
139     LFBFL_correct_sha512+='344963c62d574eb3ecb93b35aad8b5ee4c2558792478'
140     LFBFL_correct_sha512+='f7ef9da058fba1cfe2195b61c87e1ed8686f200'
141     wrapped_wget_sha512
142
143     LFBFL_file_name="create_PDF.exec.sh"
144     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
145     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
146     LFBFL_correct_sha512='e57d7603f18596a0e259dcf25f6f922cbdbad9ae34b8d'
147     LFBFL_correct_sha512+='63a3085179fcc8eb1236658a7dc81bbb72122f5e8091'
148     LFBFL_correct_sha512+='253dc270f72337ef638d83cb2490cfb187da5c1'
149     wrapped_wget_sha512
150     chmod +x "./${LFBFL_file_path}"
151
152     LFBFL_file_name="generate_from_template.libr.sh"
153     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
154     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
155     LFBFL_correct_sha512='6369d2f6f28986d56398b2a9533fddbd14251fd8c8853'
156     LFBFL_correct_sha512+='203f3d8561065b2109abc8c328f0045b4dfb47895690'
157     LFBFL_correct_sha512+='a275d560ac684aed15e41307c1531c2e88a3643'
158     wrapped_wget_sha512
159
160     LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
161     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
162     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
163     LFBFL_correct_sha512='d38b894f63ac0db1eab4447a765b723642e12e3443aee'
164     LFBFL_correct_sha512+='01bd08e44bd2971bf11d3f0b98d2932bcaaeab4f30b7'
165     LFBFL_correct_sha512+='9f7da666e02a0fb1dada8a348ad866a104e4d53'
166     wrapped_wget_sha512
167
168     LFBFL_file_name="grammar_and_spelling_check.libr.sh"
169     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
170     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
171     LFBFL_correct_sha512='48f0edc5841f59f92b8126240df13e651a887bb399640'
172     LFBFL_correct_sha512+='589fb4bdc56dfd9006e6908ad952f1ce037e6dd22293'
173     LFBFL_correct_sha512+='5667301e528a767d0ef00ad43d733990b1ae89f'
174     wrapped_wget_sha512
175
176     # /licenses_templates/ ---------------------------------------------
177     local LFBFL_dependencies_URL2
178     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
179     readonly LFBFL_dependencies_URL2
```

```
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='7af3968a0e765d5da1d15a14b3e40ba622e07ba6fa758'
LFBFL_correct_sha512+='7d3ee875e67c59da8a84d6b302affdbc8407fce7fbbc'
LFBFL_correct_sha512+='d65c52eb2965c4f1cf021f42de671f1a0739536'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
wrapped_wget_sha512

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
wrapped_wget_sha512

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
wrapped_wget_sha512
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='ec36e818b0aab4e386e4b01e643b07ad6bb1ef2ec5ccc'
LFBFL_correct_sha512+='4cfa697bac206f9a65d3cd710994a18db7397597a403'
LFBFL_correct_sha512+='463ec1353bc951ec4d79f54ccee61261dd279d8'
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='4f0aaa7e72721289d97f53e77cc735142352cb4a80bf5'
```

```
LFBFL_correct_sha512+='bc668905013a079cbca8b01d3f5bbfb7455f2c257b86'
LFBFL_correct_sha512+='6a9bf0a976c2de141995fc7871b98efaa6be217'
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2e2e9c804ec8e1879fc96df595f30994a0685934483ce'
LFBFL_correct_sha512+='c9deed4c2fb98f79d4a393e13b6511b3b740ce9f073a'
LFBFL_correct_sha512+='e7ed124f08a60b35c896d60dd710223d2592d16'
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='abe0e5701ad6a56937b1a1e3687366e87e8cb2578a179'
LFBFL_correct_sha512+='8e15ce81e0aaafc208c1b325058287c331092765ef73'
LFBFL_correct_sha512+='2c584cadf33e3acbcc195f25b0b632a0ce229d3'
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='70542f00cb59ed52d6576e6c066c5dc701ac90274ab68'
LFBFL_correct_sha512+='475086182b13b8cdf75f6b0263ac813e0010dd6cdb86'
LFBFL_correct_sha512+='9239899c2a6161f912bf1f789ad30965b8c1fe5'
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e3e8b487c45bf3960c5eb739fd1d29229ca358bfecb23'
LFBFL_correct_sha512+='e585fa7abf65c9cdaa263e799e96a83a22c660323e31'
LFBFL_correct_sha512+='bf2cdbc2f857b43957ef9405d2b8bed95dac7c0'
wrapped_wget_sha512

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='c41e25b9d06078f524ce429d92f9fb436f0e40acca954'
LFBFL_correct_sha512+='0d0b534b5bc2941f6c81bffca7ad2faaa02bf75e4ba2'
LFBFL_correct_sha512+='031bffea4e6254056474c09f18824ce6e4090cf'
wrapped_wget_sha512

LFBFL_file_name="split_score.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='0eaca0166e71a1992f897c7f87e77da3babd297cd85c6'
LFBFL_correct_sha512+='e4251ae920f15b4cbd9c3d4a7e2ae21eaf74b366a0e7'
LFBFL_correct_sha512+='681977dff58901c6d40b3e46f38e0cbec86be54'
wrapped_wget_sha512

LFBFL_file_name="split_score.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='b03772ed40ca7a90e85c52bc34715662ee94968d41801'
LFBFL_correct_sha512+='97ef5d88e10a868d68f67ed19846b522f1df9ec83d51'
LFBFL_correct_sha512+='951bf285269750de35c8316602009d1a3a75859'
wrapped_wget_sha512

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='1c5cb59a9ad306ca8cd49b398146dc6621de0e69cb510'
LFBFL_correct_sha512+='b9f41d4412d5b853f26eae7128410419be7dfb0d36cf'
LFBFL_correct_sha512+='1a6272f70f270e551406866b1e1fd209bdcce44'
wrapped_wget_sha512

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='c8bf3f61560a1f41371a913bec2d18aaa1d6c80a24b98'
LFBFL_correct_sha512+='c2580c524c3483b9a109274bb7240dca857031f686fb'
LFBFL_correct_sha512+='0b261b41b221825cc0138e2f44ed5c54acbfd20'
wrapped_wget_sha512

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='9d9100a8527ea9717aff99f483c81b7921fd079ace4f9'
LFBFL_correct_sha512+='ead5cecd3c89da90f43959297ff78184eadf55bff79e'
LFBFL_correct_sha512+='b716060befe5eff9f00cc3b2f4dc69522e8bf8f'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"
```

```
318
319     # shellcheck source=check_shell_scripts_beginnings.libr.sh
320     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
321     # shellcheck source=check_shell_scripts_indentation.libr.sh
322     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
323     # shellcheck source=check_URLs.libr.sh
324     source "./${LFBFL_subdir}/check_URLs.libr.sh"
325     # shellcheck source=common_options.libr.sh
326     source "./${LFBFL_subdir}/common_options.libr.sh"
327     # shellcheck source=comparisons.libr.sh
328     source "./${LFBFL_subdir}/comparisons.libr.sh"
329     # shellcheck source=generate_from_template.libr.sh
330     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
331     # shellcheck source=get_common_text_glob_patterns.libr.sh
332     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
333     # shellcheck source=grammar_and_spelling_check.libr.sh
334     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
335     # shellcheck source=lines_counts.libr.sh
336     source "./${LFBFL_subdir}/lines_counts.libr.sh"
337     # shellcheck source=lines_filters.libr.sh
338     source "./${LFBFL_subdir}/lines_filters.libr.sh"
339     # shellcheck source=lines_maps.libr.sh
340     source "./${LFBFL_subdir}/lines_maps.libr.sh"
341     # shellcheck source=overwrite_if_not_equal.libr.sh
342     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
343     # shellcheck source=python_black_complement.libr.sh
344     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
345     # shellcheck source=python_isort_complement.libr.sh
346     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
347     # shellcheck source=shell_checks_complement.libr.sh
348     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
349     # shellcheck source=strings_functions.libr.sh
350     source "./${LFBFL_subdir}/strings_functions.libr.sh"
351     # shellcheck source=too_long_code_lines.libr.sh
352     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
353
354     enhanced_set_bash_option extglob
355     # shellcheck source=repository_data.libr.sh
356     source "./${LFBFL_subdir}/repository_data.libr.sh"
357     # shellcheck disable=SC2154,SC2309
358     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
359       && enhanced_unset_bash_option extglob
360
361     declare -a LFBFL_return_traps_stack
362     local LFBFL_previous_return_trap
363     init_return_trap
```

```

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  --work-directory="${LFBFL_work_directory}"\
  --base-name=README\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
local LFBFL_s_files_paths
LFBFL_s_files_paths=$(
  find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
  | relevant_find
)
declare -a LFBFL_arr_files_paths
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    printf "Found template %s.\n" "${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      --work-directory="${LFBFL_some_directory}"\
      --base-name="${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done
fi

printf "Building dependencies_notes\n"
php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
  "${LFBFL_work_directory}"

declare -i LFBFL_i_directory_changed
pushd_to_work_directory
```

```
410     LFBFL_i_directory_changed=$?
411     can_continue_after_enhanced_pushd || return 1
412
413     local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"
414
415     declare -i LFBFL_i_max_line_length
416     grep_variable "${LFBFL_data_file_name}" max_line_length\
417       --result-variable-prefix=LFBFL_i_
418
419     LFBFL_some_common_options+=(
420       "--max-line-length=${LFBFL_i_max_line_length}"
421     )
422     LFBFL_some_common_options2+=(
423       "--max-line-length=${LFBFL_i_max_line_length}"
424     )
425
426     declare -i LFBFL_i_upgrade_venvs=0
427     local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
428     readonly LFBFL_upgrade_venvs_ts_file
429     declare -i LFBFL_i_upgrade_venvs_ts
430     declare -i LFBFL_i_current_ts
431     local LFBFL_upgrade_venvs_answer
432
433     local LFBFL_upgrade_venvs_time_interval_in_seconds=""
434     get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
435       "${LFBFL_some_common_options2[@]}"
436
437     if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
438       LFBFL_i_upgrade_venvs_ts=$(
439         stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
440       )
441       LFBFL_i_current_ts=$(date +%s)
442       ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
443       ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
444       if [[ LFBFL_i_current_ts -gt 0 ]]; then
445         LFBFL_i_upgrade_venvs=1
446       fi
447     else
448       LFBFL_i_upgrade_venvs=1
449     fi
450     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
451       local LFBFL_upgrade_venvs=""
452       grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
453         --result-variable-prefix=LFBFL_
454       if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
455         read -r -n 1 -t 10\
```

```
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
  }

  source_without_return_trap(){
    declare -i LFBFL_i_no_return_traps=1
    # shellcheck disable=SC1090,SC1091
    source "$1"
  }

  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
```

```
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
```

```
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
  # Saving new baseline in temp if necessary.
```

```
594   bandit --ini="${LFBFL_subdir3}/bandit.ini"\
595     --format=json\
596     --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
597     --recursive .
598   if [[ -n "${LFBFL_bandit_venv}" ]]; then
599     deactivate
600   fi
601
602   printf "Running pylint\n"
603   LFBFL_pylint_venv=""
604   grep_variable "${LFBFL_data_file_name}" pylint_venv\
605     --result-variable-prefix=LFBFL_\
606     --replace-line-returns-by=""
607   if [[ -n "${LFBFL_pylint_venv}" ]]; then
608     maybe_deactivate
609     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
610   fi
611   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
612     pip install --upgrade pylint
613   fi
614   pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
615   if [[ -n "${LFBFL_pylint_venv}" ]]; then
616     deactivate
617   fi
618
619   printf "Running ruff\n"
620   LFBFL_ruff_venv=""
621   grep_variable "${LFBFL_data_file_name}" ruff_venv\
622     --result-variable-prefix=LFBFL_\
623     --replace-line-returns-by=""
624   if [[ -n "${LFBFL_ruff_venv}" ]]; then
625     maybe_deactivate
626     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
627   fi
628   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
629     pip install --upgrade ruff
630   fi
631   ruff check --config="${LFBFL_subdir3}/ruff.toml"
632   if [[ -n "${LFBFL_ruff_venv}" ]]; then
633     deactivate
634   fi
635   printf -- "---Python end---\n"
636
637   printf -- "---PHP---\n"
638   printf "Running PHP native linter\n"
639   LFBFL_s_files_paths=$(
```

```
640       find . -type f -name "*.php"\
641       | relevant_find
642     )
643     if [[ -n "${LFBFL_s_files_paths}" ]]; then
644       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
645       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
646         php --syntax-check "${LFBFL_file_path}"
647       done
648     fi
649
650     printf "Running PHP_CodeSniffer\n"
651     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
652       composer global require squizlabs/php_codesniffer
653     fi
654     phpcs --report=code\
655       --standard=build_and_checks_variables/phpcs_ruleset.xml
656
657     printf "Running PHPMD\n"
658     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
659       composer global require phpmd/phpmd
660     fi
661
662     declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
663     local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
664     LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
665     readonly LFBFL_temp_phpmd_baseline
666
667     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
668     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
669     readonly LFBFL_phpmd_rulesets
670
671     phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
672       . text "${LFBFL_phpmd_rulesets}"
673     # Saving new baseline in temp if necessary.
674     rm "${LFBFL_temp_phpmd_baseline}"
675     phpmd --color --generate-baseline\
676       --baseline-file="${LFBFL_temp_phpmd_baseline}"\
677       . text "${LFBFL_phpmd_rulesets}"
678     sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
679     sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
680     diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
681     printf -- "---PHP end---\n"
682
683     printf -- "---JS---\n"
684     LFBFL_npm_lint_directories=""
685     grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
```

```
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©“”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

```

```
  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
```

```
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

### 3.9 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```
# the GNU General Public License
# along with DevOrSysAdminScripts.
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download

  declare -a LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    # "$--work-directory={LFBFL_work_directory}" doesn't work
    "${LFBFL_work_directory_option}"
```

```
)

source ./wget_sha512.libr.sh

local LFBFL_subdir="build_and_checks_dependencies"
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_subdir3="build_and_checks_variables"
local LFBFL_file_name
local LFBFL_script_download_URL
local LFBFL_file_path
local LFBFL_correct_sha512

wrapped_wget_sha512(){
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

LFBFL_file_name="build_md_from_printable_md.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_md_from_printable_md.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.exec.php@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_beginnings.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wrapped_wget_sha512
```

```
111     LFBFL_file_name="check_URLs.libr.sh"
112     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
113     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
114     @sha512_check_URLs.libr.sh@
115     wrapped_wget_sha512
116
117     LFBFL_file_name="common_options.libr.sh"
118     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
119     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
120     @sha512_common_options.libr.sh@
121     wrapped_wget_sha512
122
123     LFBFL_file_name="comparisons.libr.sh"
124     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
125     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
126     @sha512_comparisons.libr.sh@
127     wrapped_wget_sha512
128
129     LFBFL_file_name="create_PDF.exec.sh"
130     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
131     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
132     @sha512_create_PDF.exec.sh@
133     wrapped_wget_sha512
134     chmod +x "./${LFBFL_file_path}"
135
136     LFBFL_file_name="generate_from_template.libr.sh"
137     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
138     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
139     @sha512_generate_from_template.libr.sh@
140     wrapped_wget_sha512
141
142     LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
143     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
144     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
145     @sha512_get_common_text_glob_patterns.libr.sh@
146     wrapped_wget_sha512
147
148     LFBFL_file_name="grammar_and_spelling_check.libr.sh"
149     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
150     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
151     @sha512_grammar_and_spelling_check.libr.sh@
152     wrapped_wget_sha512
153
154     # /licenses_templates/ ---------------------------------------------
155     local LFBFL_dependencies_URL2
156     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
```

```
157     readonly LFBFL_dependencies_URL2
158     LFBFL_file_name="build_licenses_templates.exec.sh"
159     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
160     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
161     @sha512_build_licenses_templates.exec.sh@
162     wrapped_wget_sha512
163     chmod +x "./${LFBFL_file_path}"
164
165     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
166     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
167     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
168     @sha512_license_file_header_AGPLv3+.tpl@
169     wrapped_wget_sha512
170
171     LFBFL_file_name="license_file_header_GPLv3+.tpl"
172     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
173     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
174     @sha512_license_file_header_GPLv3+.tpl@
175     wrapped_wget_sha512
176
177     LFBFL_file_name="license_file_header_LGPLv3+.tpl"
178     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
179     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
180     @sha512_license_file_header_LGPLv3+.tpl@
181     wrapped_wget_sha512
182     # ------------------------------------------------------------------
183
184     LFBFL_file_name="lines_counts.libr.sh"
185     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
186     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
187     @sha512_lines_counts.libr.sh@
188     wrapped_wget_sha512
189
190     LFBFL_file_name="lines_filters.libr.sh"
191     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
192     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
193     @sha512_lines_filters.libr.sh@
194     wrapped_wget_sha512
195
196     LFBFL_file_name="lines_maps.libr.sh"
197     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
198     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
199     @sha512_lines_maps.libr.sh@
200     wrapped_wget_sha512
201
202     LFBFL_file_name="overwrite_if_not_equal.libr.sh"
```

```
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_isort_complement.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_repository_data.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_shell_checks_complement.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="split_score.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.exec.php@
wrapped_wget_sha512

LFBFL_file_name="split_score.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.libr.php@
wrapped_wget_sha512

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_strings_functions.libr.sh@
wrapped_wget_sha512
```

```
249
250     LFBFL_file_name="too_long_code_lines.libr.sh"
251     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
252     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
253     @sha512_too_long_code_lines.libr.sh@
254     wrapped_wget_sha512
255
256     LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
257     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
258     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
259     @sha512_update_or_check_files_names_listing.exec.sh@
260     wrapped_wget_sha512
261     chmod +x "./${LFBFL_file_path}"
262
263     # shellcheck source=check_shell_scripts_beginnings.libr.sh
264     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
265     # shellcheck source=check_shell_scripts_indentation.libr.sh
266     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
267     # shellcheck source=check_URLs.libr.sh
268     source "./${LFBFL_subdir}/check_URLs.libr.sh"
269     # shellcheck source=common_options.libr.sh
270     source "./${LFBFL_subdir}/common_options.libr.sh"
271     # shellcheck source=comparisons.libr.sh
272     source "./${LFBFL_subdir}/comparisons.libr.sh"
273     # shellcheck source=generate_from_template.libr.sh
274     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
275     # shellcheck source=get_common_text_glob_patterns.libr.sh
276     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
277     # shellcheck source=grammar_and_spelling_check.libr.sh
278     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
279     # shellcheck source=lines_counts.libr.sh
280     source "./${LFBFL_subdir}/lines_counts.libr.sh"
281     # shellcheck source=lines_filters.libr.sh
282     source "./${LFBFL_subdir}/lines_filters.libr.sh"
283     # shellcheck source=lines_maps.libr.sh
284     source "./${LFBFL_subdir}/lines_maps.libr.sh"
285     # shellcheck source=overwrite_if_not_equal.libr.sh
286     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
287     # shellcheck source=python_black_complement.libr.sh
288     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
289     # shellcheck source=python_isort_complement.libr.sh
290     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
291     # shellcheck source=shell_checks_complement.libr.sh
292     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
293     # shellcheck source=strings_functions.libr.sh
294     source "./${LFBFL_subdir}/strings_functions.libr.sh"
```

```
295     # shellcheck source=too_long_code_lines.libr.sh
296     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
297
298     enhanced_set_bash_option extglob
299     # shellcheck source=repository_data.libr.sh
300     source "./${LFBFL_subdir}/repository_data.libr.sh"
301     # shellcheck disable=SC2154,SC2309
302     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
303       && enhanced_unset_bash_option extglob
304
305     declare -a LFBFL_return_traps_stack
306     local LFBFL_previous_return_trap
307     init_return_trap
308
309     enhanced_set_shell_option pipefail --trap-unset
310
311     declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
312     keep_options LFBFL_some_common_options2 --verbose
313
314     printf "Building license headers\n"
315     "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
316       "${LFBFL_some_common_options[@]}"
317
318     printf "Building README.md\n"
319     "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
320       --work-directory="${LFBFL_work_directory}"\
321       --base-name=README\
322       "${LFBFL_verbose}"
323
324     printf "Building other MarkDown files\n"
325     local LFBFL_some_directory
326     declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
327     local LFBFL_s_files_paths
328     LFBFL_s_files_paths=$(
329       find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
330       | relevant_find
331     )
332     declare -a LFBFL_arr_files_paths
333     if [[ -n "${LFBFL_s_files_paths}" ]]; then
334       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
335       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
336         [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
337         printf "Found template %s.\n" "${LFBFL_file_path}"
338         LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
339         LFBFL_file_name=$(basename "${LFBFL_file_path}")
340         LFBFL_file_name=${LFBFL_file_name%.md.tpl}
```

```
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    "${LFBFL_work_directory}"

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix=LFBFL_i_

  LFBFL_some_common_options+=(
    "--max-line-length=${LFBFL_i_max_line_length}"
  )
  LFBFL_some_common_options2+=(
    "--max-line-length=${LFBFL_i_max_line_length}"
  )

  declare -i LFBFL_i_upgrade_venvs=0
  local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
  readonly LFBFL_upgrade_venvs_ts_file
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
```

```
387       ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
388       if [[ LFBFL_i_current_ts -gt 0 ]]; then
389         LFBFL_i_upgrade_venvs=1
390       fi
391     else
392       LFBFL_i_upgrade_venvs=1
393     fi
394     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
395       local LFBFL_upgrade_venvs=""
396       grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
397         --result-variable-prefix=LFBFL_
398       if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
399         read -r -n 1 -t 10\
400           -p "Upgrade venvs and composer global? [Y/n]"\
401           LFBFL_upgrade_venvs_answer
402         if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
403           LFBFL_i_upgrade_venvs=0
404         fi
405       fi
406     fi
407
408     printf "Running shellcheck\n"
409     LFBFL_s_files_paths=$(
410       find . -type f -name "*.sh"\
411       | relevant_find
412     )
413     if [[ -n "${LFBFL_s_files_paths}" ]]; then
414       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
415       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
416         shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
417           "${LFBFL_file_path}"
418       done
419     fi
420     printf "Running shell_checks_complement\n"
421     shell_checks_complement "${LFBFL_some_common_options2[@]}"
422
423     printf -- "---Python---\n"
424
425     maybe_deactivate(){
426       if command -v deactivate
427       then
428         deactivate
429       fi
430     }
431
432     source_without_return_trap(){
```

```
    declare -i LFBFL_i_no_return_traps=1
    # shellcheck disable=SC1090,SC1091
    source "$1"
  }

  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
```

```
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
```

```
525       --result-variable-prefix=LFBFL_\
526       --replace-line-returns-by=""
527     if [[ -n "${LFBFL_bandit_venv}" ]]; then
528       maybe_deactivate
529       source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
530     fi
531     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
532       pip install --upgrade bandit
533     fi
534     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
535       --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
536       --recursive .
537     # Saving new baseline in temp if necessary.
538     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
539       --format=json\
540       --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
541       --recursive .
542     if [[ -n "${LFBFL_bandit_venv}" ]]; then
543       deactivate
544     fi
545
546     printf "Running pylint\n"
547     LFBFL_pylint_venv=""
548     grep_variable "${LFBFL_data_file_name}" pylint_venv\
549       --result-variable-prefix=LFBFL_\
550       --replace-line-returns-by=""
551     if [[ -n "${LFBFL_pylint_venv}" ]]; then
552       maybe_deactivate
553       source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
554     fi
555     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
556       pip install --upgrade pylint
557     fi
558     pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
559     if [[ -n "${LFBFL_pylint_venv}" ]]; then
560       deactivate
561     fi
562
563     printf "Running ruff\n"
564     LFBFL_ruff_venv=""
565     grep_variable "${LFBFL_data_file_name}" ruff_venv\
566       --result-variable-prefix=LFBFL_\
567       --replace-line-returns-by=""
568     if [[ -n "${LFBFL_ruff_venv}" ]]; then
569       maybe_deactivate
570       source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
```

```
571     fi
572     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
573       pip install --upgrade ruff
574     fi
575     ruff check --config="${LFBFL_subdir3}/ruff.toml"
576     if [[ -n "${LFBFL_ruff_venv}" ]]; then
577       deactivate
578     fi
579     printf -- "---Python end---\n"
580
581     printf -- "---PHP---\n"
582     printf "Running PHP native linter\n"
583     LFBFL_s_files_paths=$(
584       find . -type f -name "*.php"\
585       | relevant_find
586     )
587     if [[ -n "${LFBFL_s_files_paths}" ]]; then
588       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
589       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
590         php --syntax-check "${LFBFL_file_path}"
591       done
592     fi
593
594     printf "Running PHP_CodeSniffer\n"
595     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
596       composer global require squizlabs/php_codesniffer
597     fi
598     phpcs --report=code\
599       --standard=build_and_checks_variables/phpcs_ruleset.xml
600
601     printf "Running PHPMD\n"
602     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
603       composer global require phpmd/phpmd
604     fi
605
606     declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
607     local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
608     LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
609     readonly LFBFL_temp_phpmd_baseline
610
611     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
612     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
613     readonly LFBFL_phpmd_rulesets
614
615     phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
616       . text "${LFBFL_phpmd_rulesets}"
```

```
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file="${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
  sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
```

```
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
```

```
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

### 3.10 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

# RETURN traps handling
# Add the following lines at the start of a function.
#   declare -a LFBFL_return_traps_stack
#   local LFBFL_previous_return_trap
#   init_return_trap

init_return_trap(){
  local LFBFL_previous_return_trap2
  LFBFL_previous_return_trap2=$(trap -p RETURN)
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
```

```
    get_where_was_i 2
    printf "%s previous_return_trap is: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap2}"
  fi
  if [[ -z "${LFBFL_previous_return_trap2}" ]]; then
    LFBFL_previous_return_trap2="trap - RETURN"
  fi
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  LFBFL_previous_return_trap="${LFBFL_function_depth}:"
  LFBFL_previous_return_trap+="${LFBFL_previous_return_trap2}"
  trap 'execute_return_traps' RETURN
}

execute_return_traps(){
  if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
  if [[ "${FUNCNAME[1]}" == "source" ]]; then
    # Not Working
    return
  fi
  if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  local LFBFL_some_return_trap
  local LFBFL_where_was_i
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ -n "${LFBFL_return_traps_stack}" ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "%s executing some return trap: %s.\n"\
        "${LFBFL_where_was_i}"\
        "${LFBFL_some_return_trap}"
    fi
    eval "${LFBFL_some_return_trap}"
```

```
    unset 'LFBFL_return_traps_stack[-1]'
  done
  if [[ "${LFBFL_previous_return_trap}" != ${LFBFL_function_depth}:* ]];
  then
    # We only set the previous return trap if such a trap was replaced
    # by an init_return_trap.
    return
  fi
  LFBFL_previous_return_trap=${LFBFL_previous_return_trap#*:}
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "%s setting back to previous_return_trap: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap}"
  fi
  eval "${LFBFL_previous_return_trap}"
}


is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
```

```
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd "$1" || {
    i_enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}
```

```

can_continue_after_enhanced_pushd(){
  [[ i_enhanced_pushd_result -eq 0
  || i_enhanced_pushd_result -eq 110
  || i_enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
    LFBFL_i_verbose=1
```

```
236   fi
237   # shellcheck disable=SC2034
238   readonly LFBFL_i_verbose
239 }
240
241
242 get_some_option(){
243   # $1=variable_name
244   # $2=some_option_long_name
245   # $3=option_default_value
246   # $4=read_only 0 or 1 to make the variable readonly
247   # This command is to be called in another one with same arguments on top
248   # of the three previous.
249   local LFBFL_arg
250   for LFBFL_arg in "$@"; do
251     if [[ "${LFBFL_arg}" == "$2"=* ]]; then
252       printf -v "$1" "%s" "${LFBFL_arg#"$2"=}"
253       printf -v "$1" "%s" "${!1:-$3}"
254       break
255     fi
256   done
257   if [[ $4 -eq 1 ]]; then
258     readonly "$1"
259   fi
260 }
261
262 keep_options(){
263   # $1=variable_name of array of options
264   # Other arguments: the options to keep.
265   declare -a LFBFL_array_of_filtered_options
266   declare -n LFBFL_array_of_options=$1
267   local LFBFL_some_argument
268   local LFBFL_some_argument2
269   for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
270     for LFBFL_some_argument2 in "$@"; do
271       if [[
272         "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
273         || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
274       ]]; then
275         LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
276       fi
277     done
278   done
279   LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
280 }
281
```

```
282
283 # Main usage for the next functions relative to work_directory:
284 # Add the five following lines at the start of a function.
285 #   local LFBFL_work_directory=""
286 #   get_work_directory_option "$@"
287 #   pushd_to_work_directory\
288 #     && trap 'popd_from_work_directory' RETURN
289 #   can_continue_after_enhanced_pushd || return 1
290 # Or:
291 #   local LFBFL_work_directory=""
292 #   get_work_directory_option "$@"
293 #   pushd_to_work_directory --trap-popd
294 #   can_continue_after_enhanced_pushd || return 1
295
296 get_work_directory_option(){
297   # This command is to be called in another one with same arguments.
298   # Options:
299   #   --work-directory=""
300   get_some_option LFBFL_work_directory --work-directory "." 0 "$@"
301   # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
302   if [[ -n "${LFBFL_work_directory}" ]]; then
303     LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
304   fi
305   readonly LFBFL_work_directory
306 }
307
308 pushd_to_work_directory(){
309   # Options:
310   #   --trap-popd
311   enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
312   if [[ i_enhanced_pushd_result -eq 0 ]]; then
313     local LFBFL_arg
314     for LFBFL_arg in "$@"; do
315       if [[ "${LFBFL_arg}" == --trap-popd ]]; then
316         declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
317         local LFBFL_some_return_trap
318         LFBFL_some_return_trap="${LFBFL_function_depth}:"
319         LFBFL_some_return_trap+="popd_from_work_directory 4"
320         readonly LFBFL_some_return_trap
321         LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
322         break
323       fi
324     done
325   fi
326   # shellcheck disable=SC2248
327   return ${i_enhanced_pushd_result}
```

```
}

popd_from_work_directory(){
  # $1=offset for where_was_i
  enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


get_offset_option(){
  # This command is to be called in another one with same arguments on top
  # of the default value.
  # $1=option_default_value
  # Options:
  #   --offset=2
  get_some_option LFBFL_i_offset --offset "$1" 1 "$@"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
#   or better if you use my RETURN traps handler:
#     enhanced_set_shell_option pipefail --trap-unset
#     enhanced_unset_shell_option pipefail --trap-set
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN
#   or better if you use my RETURN traps handler:
#     enhanced_set_bash_option globstar --trap-unset
#     enhanced_unset_bash_option globstar --trap-set

enhanced_set_shell_option(){
  # $1=optname some shell option for set
```

```
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
```

```
  fi
  set +o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
```

```
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
```

```
      LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

### 3.11 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
```

```
35    # l'idée d'un principe de première différence ternaire.
36    declare -i LFBFL_i_first=1
37    local LFBFL_current
38    local LFBFL_arg
39    for LFBFL_arg in "$@"; do
40      if [[ LFBFL_i_first -eq 1 ]]; then
41        LFBFL_current="${LFBFL_arg}"
42        LFBFL_i_first=0
43        continue
44      fi
45      if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
46        return 0
47      fi
48    done
49    return 1
50  }
51
52  some_distinct(){
53    ! equal "$@"
54  }
55
56  all_distinct(){
57    # Returns 1 if all the arguments strings are distinct.
58    declare -ir LFBFL_ij_max=$#
59    declare -i LFBFL_i
60    declare -i LFBFL_j
61    for ((LFBFL_i=1; LFBFL_i<LFBFL_ij_max; ++LFBFL_i)); do
62      for ((LFBFL_j = LFBFL_i + 1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
63        [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
64      done
65    done
66    return 1
67  }
68
69  all_distinct2(){
70    # Returns 1 if all the arguments strings are distinct.
71    local LFBFL_element
72    local LFBFL_previous_element
73    if [[ $# -le 1 ]]; then
74      return 1
75    fi
76    declare -r LFBFL_s_sorted_arguments=$(
77      printf "%s\n" "$@" \
78      | sort
79    )
80    declare -a LFBFL_arr_sorted_arguments
```

```
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i=1; LFBFL_i<LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq -d
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

```

```
some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1"\
    | head -1
}

is_substring(){
  # $1=string
```

```
  # $2=substring
  declare -r LFBFL_var_1=$(
    printf "%s" "$1"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    printf "%s" "$2"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
  )
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php -r "${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=file_path
  # $2=sub_file_path
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

### 3.12 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# DevOrSysAdminScripts is distributed in the hope
```

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# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

```

```
58    declare -a LFBFL_common_options=("$@")
59    keep_options LFBFL_common_options --verbose
60
61    local LFBFL_subdir2="build_and_checks_variables"
62    local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
63    local LFBFL_repository_name=""
64    grep_variable "${LFBFL_data_file_name}" repository_name\
65      --result-variable-prefix="LFBFL_"\
66      --replace-line-returns-by=""
67
68    local LFBFL_repository_web_url=""
69    grep_variable "${LFBFL_data_file_name}" repository_web_url\
70      --result-variable-prefix="LFBFL_"\
71      --replace-line-returns-by=""
72
73    local LFBFL_repository_git_url=""
74    grep_variable "${LFBFL_data_file_name}" repository_git_url\
75      --result-variable-prefix="LFBFL_"\
76      --replace-line-returns-by=""
77
78    local LFBFL_abstract=""
79    grep_variable "${LFBFL_data_file_name}" abstract\
80      --result-variable-prefix="LFBFL_"
81
82    local LFBFL_acknowledgments=""
83    grep_variable "${LFBFL_data_file_name}" acknowledgments\
84      --result-variable-prefix="LFBFL_"
85
86    local LFBFL_author_email=""
87    grep_variable "${LFBFL_data_file_name}" author_email\
88      --result-variable-prefix="LFBFL_"\
89      --replace-line-returns-by=""
90
91    local LFBFL_author_full_name=""
92    grep_variable "${LFBFL_data_file_name}" author_full_name\
93      --result-variable-prefix="LFBFL_"\
94      --replace-line-returns-by=" "
95
96    local LFBFL_author_website=""
97    grep_variable "${LFBFL_data_file_name}" author_website\
98      --result-variable-prefix="LFBFL_"\
99      --replace-line-returns-by=""
100
101   declare -i LFBFL_i_max_line_length
102   grep_variable "${LFBFL_data_file_name}" max_line_length\
103     --result-variable-prefix="LFBFL_i_"
```

```
104
105   declare -r LFBFL_current_date=$(date -I"date")
106
107   declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
108
109   declare -ir LFBFL_i_number_of_commits=$(
110     git shortlog\
111     | space_starting_lines\
112     | wc -l
113   )
114
115   local LFBFL_redirect="/dev/stdout"
116   if [[ LFBFL_i_verbose -eq 0 ]]; then
117     LFBFL_redirect="/dev/null"
118   fi
119   {
120     code_lines_count_all "${LFBFL_common_options[@]}"
121     code_lines_count_not_empty "${LFBFL_common_options[@]}"
122     code_lines_count_empty "${LFBFL_common_options[@]}"
123   } &> "${LFBFL_redirect}"
124
125   local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
126   LFBFL_number_of_lines+=" total lines,"
127   LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
128   LFBFL_number_of_lines+=" not empty lines,"
129   LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
130   LFBFL_number_of_lines+=" empty lines."
131   readonly LFBFL_number_of_lines
132
133   declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"
134
135   # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
136   # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
137   # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
138   tree -a --gitignore\
139     -I "node_modules/"\
140     -I "__pycache__/"\
141     -I ".mypy_cache/"\
142     -I ".ruff_cache/"\
143     -I ".git/"\
144     | replace_non_ascii_spaces\
145     > "${LFBFL_temp_path}/current_tree_light.txt"
146
147   tree -a -DFh --gitignore\
148     -I "node_modules/"\
149     -I "__pycache__/"\
```

```
150       -I ".mypy_cache/"\
151       -I ".ruff_cache/"\
152       -I ".git/"\
153       | replace_non_ascii_spaces\
154       > "${LFBFL_temp_path}/current_tree.txt.temp"
155
156     local LFBFL_s_files_paths
157     # Remove line returns that are here to keep lines short.
158     LFBFL_s_files_paths=$(
159       sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_subdir2}/files_names_listing.txt"\
160       | grep -v '^//'
161     )
162     readonly LFBFL_s_files_paths
163     declare -a LFBFL_arr_files_paths
164     if [[ -n "${LFBFL_s_files_paths}" ]]; then
165       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
166     fi
167     readonly LFBFL_arr_files_paths
168
169     local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
170     LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
171     readonly LFBFL_temp_files_listing
172     : > "${LFBFL_temp_files_listing}"
173     local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
174     LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
175     readonly LFBFL_temp_files_listing2
176     : > "${LFBFL_temp_files_listing2}"
177     # HTML <li> elements, hence "lis".
178     local LFBFL_temp_files_lis
179     LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
180     readonly LFBFL_temp_files_lis
181     : > "${LFBFL_temp_files_lis}"
182     # shellcheck disable=SC2248
183     get_split_score_simple 1 ${LFBFL_i_max_line_length} /
184     declare -r LFBFL_score_command="${get_split_score_result}"
185     declare -i LFBFL_i_score_command_properties
186     LFBFL_i_score_command_properties=${i_get_split_score_result2}
187     readonly LFBFL_i_score_command_properties
188     # shellcheck disable=SC2248
189     get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
190     declare -r LFBFL_score_command2="${get_split_score_result}"
191     declare -i LFBFL_i_score_command_properties2
192     LFBFL_i_score_command_properties2=${i_get_split_score_result2}
193     readonly LFBFL_i_score_command_properties2
194     declare -r LFBFL_suffix='%'
195     declare -i LFBFL_i=0
```

```
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  local LFBFL_s_format
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        ${LFBFL_i_score_command_properties2}
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
```

```
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_i_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
```

```
  "${LFBFL_temp_files_listing}"
# Example of generated content of files_listing.tex.sub:
# ----------------------------------------------------------------------
# \subsection{
#   build\_and\_checks\_dependencies/licenses\_templates/%
# build\_licenses\_templates.exec.sh
# }
# \label{
#   build_and_checks_dependencies:licenses_templates:%
# build_licenses_templatesexecsh
# }
#
# \VerbatimInput[numbers=left,xleftmargin=-5mm]{
# build_and_checks_dependencies/licenses_templates/%
# build_licenses_templates.exec.sh
# }
# ----------------------------------------------------------------------
sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
  "${LFBFL_temp_files_listing2}"
# Example of generated content of files_listing.html.sub:
# ----------------------------------------------------------------------
# <h3 id="subsection3.15">3.15
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
# </h3>
# ...
# <h3 id="subsection3.33">3.33
# build_and_checks_variables/.gitignore
# </h3>
# <pre class="numbered_lines">
# temp/
# upgrade_venvs_ts
# </pre>
# ----------------------------------------------------------------------
overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
  "${LFBFL_temp_files_lis}"
# Example of generated content of files_lis.html.sub:
# ----------------------------------------------------------------------
#       <li><a href="#subsection3.15">
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
#       </a></li>
# ----------------------------------------------------------------------

# We verify if some lines are beyond max_line_length characters
```

```
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_i_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
  local LFBFL_file_name
  local LFBFL_s_lines
  declare -a LFBFL_arr_lines
  declare -i LFBFL_i_keep_length
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
    # shellcheck disable=SC2248
    LFBFL_s_lines=$(
      grep '.\{'${LFBFL_i_overlength}'\}' "${LFBFL_tree_path}"
    )
    if [[ -z "${LFBFL_s_lines}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E -e 's/(.*)──[^─]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
```

```
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E 's|.* ([^ ]+)$|\1|g'
      )
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
      LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_i_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
    # If there is a template, we init the process from it.
    cp "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
```

```
elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
  # Otherwise if there is a tex file, we init the process from it.
  cp "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
    "${LFBFL_tex_path_start}.1"
else
  printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
fi

# Same logic with repository HTML file.
if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
  cp "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
    "${LFBFL_html_path_start}.1"
elif [[ -f "${LFBFL_repository_name}.html" ]]; then
  cp "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
else
  printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
    "${LFBFL_subdir2}"
fi

# shellcheck disable=SC2001
printf "%s\n" "${LFBFL_abstract}"\
  | sed -e 's/\\n/\n/g'\
  > "${LFBFL_temp_path}/abstract_temp"

# shellcheck disable=SC2001
printf "%s\n" "${LFBFL_acknowledgments}"\
  | sed -e 's/\\n/\n/g'\
  > "${LFBFL_temp_path}/acknowledgments_temp"

# But the filling still occurs, in case the dev want to refill
# part of the tex/html file that he modified with @token@
# using some computed results.
# Tex filling:
if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
  sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
         -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
         -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
         -e "s|@author_email@|${LFBFL_author_email}|g"\
         -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
         -e "s|@author_website@|${LFBFL_author_website}|g"\
         -e "s|@current_date@|${LFBFL_current_date}|g"\
         -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
         -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
         -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
    "${LFBFL_tex_path_start}.1"

```

```
    insert_file_at_token "${LFBFL_tex_path_start}.1"\
      @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_tex_path_start}.3"

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_tex_path_start}.4"

    insert_file_at_token "${LFBFL_tex_path_start}.4"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_tex_path_start}.5"

    insert_file_at_token "${LFBFL_tex_path_start}.5"\
      @dependencies_notes@\
      "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
      "${LFBFL_tex_path_start}.6"

    insert_file_at_token "${LFBFL_tex_path_start}.6"\
      @files_listing_VerbatimInput@\
      "${LFBFL_temp_path}/files_listing.tex.sub"\
      "${LFBFL_tex_path_start}.7"

    overwrite_if_not_equal\
      "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.7" 1
  fi

  # HTML filling:
  declare -i LFBFL_i_HTML_updated=0
  if [[ -f "${LFBFL_html_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
```

```
518             -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
519             -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
520       "${LFBFL_html_path_start}.1"
521
522     insert_file_at_token "${LFBFL_html_path_start}.1"\
523       @abstract@\
524       "${LFBFL_temp_path}/abstract_temp"\
525       "${LFBFL_html_path_start}.2"
526
527     insert_file_at_token "${LFBFL_html_path_start}.2"\
528       @acknowledgments@\
529       "${LFBFL_temp_path}/acknowledgments_temp"\
530       "${LFBFL_html_path_start}.3"
531
532     insert_file_at_token "${LFBFL_html_path_start}.3"\
533       @files_lis@\
534       "${LFBFL_temp_path}/files_lis.html.sub"\
535       "${LFBFL_html_path_start}.4"
536
537     insert_file_at_token "${LFBFL_html_path_start}.4"\
538       @current_tree_light@\
539       "${LFBFL_temp_path}/current_tree_light.txt"\
540       "${LFBFL_html_path_start}.5"
541
542     insert_file_at_token "${LFBFL_html_path_start}.5"\
543       @current_tree@\
544       "${LFBFL_temp_path}/current_tree.txt"\
545       "${LFBFL_html_path_start}.6"
546
547     insert_file_at_token "${LFBFL_html_path_start}.6"\
548       @dependencies_notes@\
549       "${LFBFL_temp_path}/dependencies_notes.html.sub"\
550       "${LFBFL_html_path_start}.7"
551
552     insert_file_at_token "${LFBFL_html_path_start}.7"\
553       @files_listing_HTMLPreInput@\
554       "${LFBFL_temp_path}/files_listing.html.sub"\
555       "${LFBFL_html_path_start}.8"
556
557     overwrite_if_not_equal "${LFBFL_repository_name}.html"\
558       "${LFBFL_html_path_start}.8" 1
559     LFBFL_i_HTML_updated=$?
560   fi
561
562   # The HTML contains everything including listings with source code.
563   # If it doesn't get updated,
```

```
  # then it is useless to recompute the PDF.
  if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
    return
  fi

  local LFBFL_latex_command=""
  grep_variable "${LFBFL_data_file_name}" latex_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  case ${LFBFL_latex_command} in
    pdflatex) : ;;
    lualatex) : ;;
    *) printf "Unsupported latex command" ; return 1 ;;
  esac

  local LFBFL_command
  LFBFL_command="${LFBFL_latex_command}"
  LFBFL_command+=" '${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_command+=" > /dev/null"
  fi
  readonly LFBFL_command

  for ((i=0; i<3; i++)); do
    eval "${LFBFL_command}"
  done

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

### 3.13 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
```

```
50    # $1=base_file_path
51    # $2=target_file_path
52    # $3=line_comment_prefix
53    # $4=optional_post_processing
54    # $5=file_prefix
55    local LFBFL_temp
56    LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
57    readonly LFBFL_temp
58    if [[ -n "$5" ]]; then
59      printf "%s\n" "$5"\
60        > "${LFBFL_temp}"
61    fi
62    sed -e "s/^/$3/g" "$1"\
63      >> "${LFBFL_temp}"
64    if [[ -n "$4" ]]; then
65      eval "$4"
66    fi
67    overwrite_if_not_equal "$2" "${LFBFL_temp}"
68  }
69
70  split_file_in_two(){
71    # $1=file_path
72    # $2=token assume that token is the only thing on his line,
73    #    except for a possible indentation.
74    # $3=path_of_part1_of_file
75    # $4=path_of_part2_of_file
76    # We cannot add --line-regexp with --fixed-strings because of the
77    # possible indentation in file.
78    declare -ir LFBFL_i_line_number=$(
79      grep_fixed_string_with_anchor "$1" "$2"\
80        --enforce-line-ends-with-fixed-string --fixed-strings --line-number\
81      | head --lines=1\
82      | cut -f 1 -d ':'
83    )
84    declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
85    declare -ir LFBFL_i_lines_after=$((
86      LFBFL_i_line_count - LFBFL_i_line_number
87    ))
88    head --lines="$((LFBFL_i_line_number - 1))" "$1"\
89      > "$3"
90    tail --lines="${LFBFL_i_lines_after}" "$1"\
91      > "$4"
92  }
93
94  insert_file_at_token(){
95    # $1=file_path
```

```
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_file_to_insert
  # $4=path_of_result_file
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}"
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

### 3.14 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
# along with DevOrSysAdminScripts.
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.sub'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
```

```
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.sub'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

### 3.15 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# or (at your option) any later version.
#
```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spelling_check(){
  # $1=configuration_path build_and_checks_variables/repository_data.txt
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
```

```
  local LFBFL_grammar_and_spelling_checker_command=""
  grep_variable "$1" grammar_and_spelling_checker_command\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: No command in configuration.\n"
    fi
    return
  fi
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "grammar_and_spelling_check: Command found: %s.\n"\
      "${LFBFL_command}"
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

### 3.16 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
49    declare -a LFBFL_return_traps_stack
50    local LFBFL_previous_return_trap
51    init_return_trap
52
53    enhanced_set_shell_option pipefail --trap-unset
54
55    local LFBFL_s_format
56
57    # Source
58    local LFBFL_license_source_subdir
59    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
60    readonly LFBFL_license_source_subdir
61    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
62    LFBFL_source_license_prefix+="license_file_header_"
63    readonly LFBFL_source_license_prefix
64    # Target
65    local LFBFL_license_target_subdir
66    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
67    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
68    readonly LFBFL_license_target_subdir
69    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
70    LFBFL_target_license_prefix+="license_file_header_"
71    readonly LFBFL_target_license_prefix
72
73    declare -ar LFBFL_licenses=(
74      "AGPLv3+"
75      "GPLv3+"
76      "LGPLv3+"
77    )
78
79    # ------------------------------------------------------------------
80    # Various language definitions for text file formats
81    declare -ar LFBFL_block_comment_languages=(
82      "c"
83      "py"
84    )
85    declare -ar LFBFL_block_comment_enters=(
86      '/*'
87      '"""'
88    )
89    declare -ar LFBFL_block_comment_exits=(
90      '*/'
91      '"""'
92    )
93    equal "${#LFBFL_block_comment_languages[@]}"\
94          "${#LFBFL_block_comment_enters[@]}"\
```

```
        "${#LFBFL_block_comment_exits[@]}"
  declare -i LFBFL_i_equal_result=$?
  if [[ LFBFL_i_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
    return 1
  fi

  declare -ar LFBFL_line_comment_languages=(
    "sh"
    "sql"
    "tex"
  )
  declare -ar LFBFL_line_comment_prefixes=(
    '# '
    '-- '
    '% '
  )
  equal "${#LFBFL_line_comment_languages[@]}"\
        "${#LFBFL_line_comment_prefixes[@]}"
  LFBFL_i_equal_result=$?
  if [[ LFBFL_i_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
    return 1
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # We generate the license header templates adapted to various text
  # files formats.
  # Kamoulox do endfor, Jacques Beauregard XD
  # (C'est toujours mieux que l'inverse :) !)
  # for do endfor -> for do done (Fort Doux Donne).
  local LFBFL_some_license
  local LFBFL_license_prefix
  local LFBFL_license_prefix2
  local LFBFL_extension
  local LFBFL_license_file_name
  local LFBFL_enter_string
  local LFBFL_exit_string
  local LFBFL_temp2
  local LFBFL_prefix_string
  local LFBFL_intermediate_file_name
  local LFBFL_file_prefix
  declare -i LFBFL_i_generate_from_template_result
  declare -i LFBFL_i
  for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
```

```
141     LFBFL_license_prefix="${LFBFL_source_license_prefix}"
142     LFBFL_license_prefix+="${LFBFL_some_license}"
143     LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
144     LFBFL_license_prefix2+="${LFBFL_some_license}"
145     for ((
146       LFBFL_i = 0;
147       LFBFL_i < ${#LFBFL_block_comment_languages[@]};
148       ++LFBFL_i
149     )); do
150       LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
151       LFBFL_license_file_name="${LFBFL_license_prefix2}"
152       LFBFL_license_file_name+=".${LFBFL_extension}"
153       LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
154       LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
155       generate_from_template_with_block_comments\
156         "${LFBFL_license_prefix}.tpl"\
157         "${LFBFL_license_file_name}.tpl"\
158         "${LFBFL_enter_string}" "${LFBFL_exit_string}"
159       LFBFL_i_generate_from_template_result=$?
160       if [[ LFBFL_i_verbose -eq 1 ]]; then
161         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
162           printf "License template %s.tpl updated.\n"\
163             "${LFBFL_license_file_name}"
164         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
165           printf "License template %s.tpl created.\n"\
166             "${LFBFL_license_file_name}"
167         fi
168       fi
169     done
170     LFBFL_temp2=".generate_from_template_with_line_comments.temp"
171     for ((
172       LFBFL_i = 0;
173       LFBFL_i < ${#LFBFL_line_comment_languages[@]};
174       ++LFBFL_i
175     )); do
176       LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
177       LFBFL_license_file_name="${LFBFL_license_prefix2}"
178       LFBFL_license_file_name+=".${LFBFL_extension}"
179       LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
180       LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
181       LFBFL_intermediate_file_name+="${LFBFL_temp2}"
182       LFBFL_file_prefix=""
183       if [[ "${LFBFL_extension}" == "sh" ]]; then
184         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
185       fi
186       # The sed command is here to remove trailing spaces of previously
```

```
      # empty lines after comment prefix is added to the line.
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
        "${LFBFL_file_prefix}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_name}"
        fi
      fi
    done
  done
  # ---------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in work directory, we can go there.
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="build_and_checks_variables/"
  LFBFL_data_file_name+="repository_data.txt"
  readonly LFBFL_data_file_name
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_name}" license\
    --result-variable-prefix="LFBFL_"
  local LFBFL_license2=""
  grep_variable "${LFBFL_data_file_name}" license2\
    --result-variable-prefix="LFBFL_"
  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  LFBFL_license_prefix="build_and_checks_variables/temp/"
  LFBFL_license_prefix+="license_file_header_"
```

```
233   readonly LFBFL_license_prefix
234   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
235   readonly LFBFL_license_prefix2
236   declare -r\
237     LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
238
239   # First year according to current state of git repository.
240   declare -ir LFBFL_i_first_year=$(
241     git log\
242     | grep 'Date:'\
243     | tail -1\
244     | cut -f 8 -d ' '
245   )
246   # Last year according to current state of git repository.
247   declare -ir LFBFL_i_last_year=$(
248     git log\
249     | grep 'Date:'\
250     | head -1\
251     | cut -f 8 -d ' '
252   )
253   local LFBFL_copyright_string
254   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
255   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
256   readonly LFBFL_copyright_string
257
258   prepare_filled_license_file_block(){
259     # $1=license_file_name
260     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
261         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
262         "$1.tpl"\
263       > "$1.temp"
264     overwrite_if_not_equal "$1" "$1.temp"
265     declare -ir LFBFL_i_result=$?
266     if [[ LFBFL_i_verbose -eq 1 ]]; then
267       if [[ LFBFL_i_result -eq 1 ]]; then
268         printf "License template %s updated.\n" "$1"
269       elif [[ LFBFL_i_result -eq 2 ]]; then
270         printf "License template %s created.\n" "$1"
271       fi
272     fi
273     head --lines=-1 "$1"\
274       | tail --lines=+2\
275       > "$1.temp"
276     # Strange world: tail --lines=-1 was working
277     # to exclude first line.
278     # I tested it months ago, and saw with my eyes the start and end
```

```
    # block comments removed with:
    # head --lines=-1 "$1" | tail --lines=-1.
    # Now, man page says to use --lines=+2 instead.
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
    ["js"]="c"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_name2
  declare -i LFBFL_i_not_subfile
  declare -i LFBFL_i_not_subfile2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_name2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
```

```
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}"\
        "${LFBFL_license_file_name}.temp"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}"\
          "${LFBFL_license_file_name2}.temp"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
    rm "${LFBFL_license_file_name}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
      rm "${LFBFL_license_file_name2}.temp"
    fi
  done

  prepare_filled_license_file_line(){
    # $1=LFBFL_license_file_name
    sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
        -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
        "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -ir LFBFL_i_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_i_result -eq 1 ]]; then
        printf "License template %s updated.\n" "$1"
      elif [[ LFBFL_i_result -eq 2 ]]; then
        printf "License template %s created.\n" "$1"
      fi
    fi
  }

  declare -Ar LFBFL_all_line_comment_languages=(
    ["sh"]="sh"
    ["sql"]="sql"
    ["tex"]="tex"
  )

```

```
  for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_line "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_line "${LFBFL_license_file_name2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name}"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name2}"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

### 3.17 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

### 3.18 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

```
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```

```
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```

### 3.19 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

### 3.20 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
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#
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "all_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep --invert-match --perl-regexp --with-filename 'a(?!a)a'
  done
}
```

```
62
63 all_self_code_lines(){
64   # Options:
65   #   --verbose
66   #   --work-directory=""
67   declare -i LFBFL_i_verbose=0
68   get_verbose_option "$@"
69   local LFBFL_work_directory=""
70   get_work_directory_option "$@"
71   declare -a LFBFL_return_traps_stack
72   local LFBFL_previous_return_trap
73   init_return_trap
74   pushd_to_work_directory --trap-popd
75   can_continue_after_enhanced_pushd || return 1
76
77   enhanced_set_shell_option pipefail --trap-unset
78
79   declare -a LFBFL_common_options=("$@")
80   keep_options LFBFL_common_options --verbose
81
82   # below and another lines, see SC2145 false positive
83   all_code_lines "${LFBFL_common_options[@]}" \
84     | relevant_grep\
85     | not_license_grep\
86     | not_main_tex_grep\
87     | not_main_html_grep
88 }
89
90 # shellcheck disable=SC2120
91 all_self_empty_code_lines(){
92   # Options:
93   #   --verbose
94   #   --work-directory=""
95   declare -i LFBFL_i_verbose=0
96   get_verbose_option "$@"
97   local LFBFL_work_directory=""
98   get_work_directory_option "$@"
99   declare -a LFBFL_return_traps_stack
100  local LFBFL_previous_return_trap
101  init_return_trap
102  pushd_to_work_directory --trap-popd
103  can_continue_after_enhanced_pushd || return 1
104
105  enhanced_set_shell_option pipefail --trap-unset
106
107  declare -a LFBFL_common_options=("$@")
```

```
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

```

```
154   declare -gi i_code_lines_count_all_result
155   i_code_lines_count_all_result=$(
156     all_self_code_lines\
157     | wc --lines
158   )
159   # The printf is for command line use.
160   # A script should use the result value instead.
161   # shellcheck disable=SC2248
162   printf "%s\n" ${i_code_lines_count_all_result}
163 }
164
165 code_lines_count_empty(){
166   # Options:
167   #   --verbose
168   #   --work-directory=""
169   declare -i LFBFL_i_verbose=0
170   get_verbose_option "$@"
171   local LFBFL_work_directory=""
172   get_work_directory_option "$@"
173   declare -a LFBFL_return_traps_stack
174   local LFBFL_previous_return_trap
175   init_return_trap
176   pushd_to_work_directory --trap-popd
177   can_continue_after_enhanced_pushd || return 1
178
179   enhanced_set_shell_option pipefail --trap-unset
180
181   declare -gi i_code_lines_count_empty_result
182   # shellcheck disable=SC2119
183   i_code_lines_count_empty_result=$(
184     all_self_empty_code_lines\
185     | wc --lines
186   )
187   # The printf is for command line use.
188   # A script should use the result value instead.
189   # shellcheck disable=SC2248
190   printf "%s\n" ${i_code_lines_count_empty_result}
191 }
192
193 code_lines_count_not_empty(){
194   # Options:
195   #   --verbose
196   #   --work-directory=""
197   declare -i LFBFL_i_verbose=0
198   get_verbose_option "$@"
199   local LFBFL_work_directory=""
```

```
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

### 3.21 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
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```

```
# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
```

```
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41
```

```
116
117 in_place_grep(){
118   # Same arguments and options as grep.
119   declare -r LFBFL_temp="${!#}.in_place_grep.temp"
120   grep "$@" \
121     > "${LFBFL_temp}"
122   # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
123   # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
124   # printf "%s\n" "${LFBFL_i_lines_before}"
125   # printf "%s\n" "${LFBFL_i_lines_after}"
126   # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
127   # Almost as fast (on large files) and correct in all cases :).
128   # Thanks Paul Eggert for the suggestion :).
129   # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
130   if cmp --silent "${LFBFL_temp}" "${!#}"; then
131     rm "${LFBFL_temp}"
132     return
133   fi
134   mv "${LFBFL_temp}" "${!#}"
135 }
136
137 grep_fixed_string_with_anchor(){
138   # $1=file_path
139   # $2=some_fixed_string
140   # Options:
141   #   --enforce-line-starts-with-fixed-string
142   #   --enforce-line-ends-with-fixed-string
143   #   Apart from $1 and $2 and the options above,
144   #   all other arguments are considered grep_options like --quiet
145   #   for example.
146   #   But note that --fixed-strings will be added below, you don't need to
147   #   supply it.
148   declare -a LFBFL_grep_options=()
149   declare -a LFBFL_grep_options2=()
150   declare -i LFBFL_i=0
151   declare -i LFBFL_i_enforce_line_starts=0
152   declare -i LFBFL_i_enforce_line_ends=0
153   declare -i LFBFL_i_quiet=0
154   local LFBFL_arg
155   for LFBFL_arg in "$@"; do
156     ((++LFBFL_i))
157     if [[ LFBFL_i -lt 3 ]]; then
158       continue
159     fi
160     if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
161     then
```

```
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]]; then
      LFBFL_i_quiet=1
      LFBFL_grep_options+=("${LFBFL_arg}")
      continue
    fi
    LFBFL_grep_options2+=("${LFBFL_arg}")
  done

  if [[
    LFBFL_i_enforce_line_starts -eq 0
    && LFBFL_i_enforce_line_ends -eq 0
  ]]; then
    grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
    return
  fi
  declare -r LFBFL_s_lines=$(
    grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
      "$2" "$1"
  )
  declare -i LFBFL_i_result=1
  if [[ -z "${LFBFL_s_lines}" ]]; then
    # shellcheck disable=SC2248
    return ${LFBFL_i_result}
  fi
  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  declare -i LFBFL_i_fixed_string_length=${#2}
  local LFBFL_some_line
  local LFBFL_line_part
  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "$2" ]]; then
      LFBFL_i_result=0
      if [[ LFBFL_i_quiet -eq 1 ]]; then
        break
      else
        printf "%s\n" "${LFBFL_some_line}"
```

```
208         fi
209         continue
210       fi
211       if [[
212         LFBFL_i_enforce_line_starts -eq 1
213         && LFBFL_i_enforce_line_ends -eq 1
214       ]]; then
215         continue
216       fi
217       if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
218         LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
219       fi
220       if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
221         LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
222       fi
223       if [[ "${LFBFL_line_part}" == "$2" ]]; then
224         LFBFL_i_result=0
225         if [[ LFBFL_i_quiet -eq 1 ]]; then
226           break
227         else
228           printf "%s\n" "${LFBFL_some_line}"
229         fi
230       fi
231     done
232     LFBFL_i_result=$((LFBFL_i_result | $?))
233     # shellcheck disable=SC2248
234     return ${LFBFL_i_result}
235   }
236
237   grep_variable(){
238     # $1=file_path
239     # $2=variable_name
240     # Options :
241     #   --replace-line-returns-by=""
242     #   --result-variable-prefix="LFBFL_" for example
243     declare -r LFBFL_regexp="(?<=^$2=).*$"
244     local LFBFL_arg
245     # printf "%s\n" "${LFBFL_regexp}"
246     local LFBFL_variable_value
247     if [[ "$*" == *--replace-line-returns-by=* ]]; then
248       local LFBFL_s_replace=""
249       for LFBFL_arg in "$@"; do
250         if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
251           LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
252           break
253         fi
```

```
    done
    # sed -Ez -e...
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"\
      | sed --null-data --regexp-extended\
        --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"
    )
  fi
  local LFBFL_prefix=""
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep --invert-match '^$'
}

space_starting_lines(){
  grep '^ '
}

```

```
not_space_starting_lines(){
  grep --invert-match '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep --extended-regexp '^( |$)'
}

not_spampty_lines(){
  grep '^[^ ]'
  # grep --extended-regexp --invert-match '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep --invert-match '^[ ]\+$'
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep --invert-match '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep --invert-match '^[^:]\+: '
```

```
}

spampty_lines_after_file_name(){
  grep --extended-regexp '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep --extended-regexp --invert-match '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# -E = --show-ends :). Nice command :).

not_JS_dependencies_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)node_modules/"\
    --regexp="(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep --extended-regexp --invert-match\
```

```
    --regexp="^([^:]+/)?node_modules/"\
    --regexp="^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_php_cache_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)\.phpdoc/cache/"
}

not_php_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?\.phpdoc/cache/"
}

not_python_cache_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)__pycache__/"\
    --regexp="(^|/)\.mypy_cache/"\
    --regexp="(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?__pycache__/"\
    --regexp="^([^:]+/)?\.mypy_cache/"\
    --regexp="^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_php_cache_find\
    | not_python_cache_find
}

not_cache_grep(){
  not_php_cache_grep\
    | not_python_cache_grep
}
```

```
not_git_find(){
  grep --extended-regexp --invert-match "(^|/)\.git/"
}

not_git_grep(){
  grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
```

```
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)build_and_checks_variables/temp/"
}

not_temp_file_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}

not_unchecked_builds_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)\.phpdoc/build/"
}

not_unchecked_builds_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?\.phpdoc/build/"
}

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
```

```
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find\
    | not_unchecked_builds_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep\
    | not_unchecked_builds_grep
}
```

### 3.22 build_and_checks_dependencies/lines_maps.libr.sh

```
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

```

```
# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
      -e 's/\xe2\x80\xab/ /g'\
      -e 's/\xe2\x80\xac/ /g'\
      -e 's/\xe2\x80\xad/ /g'\
      -e 's/\xe2\x80\xae/ /g'\
      -e 's/\xe2\x80\xaf/ /g'\
      -e 's/\xe2\x81\x9f/ /g'\
      -e 's/\xe2\x81\xa0/ /g'\
      -e 's/\xe2\x81\xa1/ /g'\
      -e 's/\xe2\x81\xa2/ /g'\
      -e 's/\xe2\x81\xa3/ /g'\
      -e 's/\xe2\x81\xa4/ /g'\
      -e 's/\xe2\x81\xa5/ /g'\
      -e 's/\xe2\x81\xa6/ /g'\
      -e 's/\xe2\x81\xa7/ /g'\
```

```
      -e 's/\xe2\x81\xa8/ /g'\
      -e 's/\xe2\x81\xa9/ /g'\
      -e 's/\xe2\x81\xaa/ /g'\
      -e 's/\xe2\x81\xab/ /g'\
      -e 's/\xe2\x81\xac/ /g'\
      -e 's/\xe2\x81\xad/ /g'\
      -e 's/\xe2\x81\xae/ /g'\
      -e 's/\xe2\x81\xaf/ /g'
}
```

### 3.23 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # $3=keep_temp
  # $4=tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories timestamps.
  # $5=--quiet : we even don't want the message of diff saying that
  #              files are distinct
```

```
35    # Return 0 target was the same
36    #        1 target was different
37    #        2 target did not already exist
38    if [[ ! -f "$1" ]]; then
39      if [[ -n "$3" ]]; then
40        cp -p "$2" "$1"
41      else
42        mv "$2" "$1"
43      fi
44      return 2
45    fi
46    declare -i LFBFL_i_is_equal
47    if [[ -n "$4" ]]; then
48      # Not setting pipefail since the result would still be incorrect.
49      # shellcheck disable=SC2312
50      diff "$1" "$2"\
51        | grep --extended-regexp "^(>|<)"\
52        | grep --invert-match "/$"
53      LFBFL_i_is_equal=${PIPESTATUS[2]}
54      if [[ LFBFL_i_is_equal -eq 1 ]]; then
55        grep --only-matching "[^ ]*/$" "$1"\
56          > "overwrite_if_not_equal.file1.temp"
57        grep --only-matching "[^ ]*/$" "$2"\
58          > "overwrite_if_not_equal.file2.temp"
59        if [[ -n "$5" ]]; then
60          diff "overwrite_if_not_equal.file1.temp"\
61            "overwrite_if_not_equal.file2.temp"\
62            > /dev/null
63        else
64          diff "overwrite_if_not_equal.file1.temp"\
65            "overwrite_if_not_equal.file2.temp"
66        fi
67        LFBFL_i_is_equal=1-$?
68        rm "overwrite_if_not_equal.file1.temp"\
69          "overwrite_if_not_equal.file2.temp"
70      fi
71    else
72      if [[ -n "$5" ]]; then
73        diff --brief "$1" "$2" > /dev/null
74      else
75        diff --brief "$1" "$2"
76      fi
77      LFBFL_i_is_equal=1-$?
78    fi
79    if [[ LFBFL_i_is_equal -eq 1 ]]; then
80      if [[ -z "$3" ]]; then
```

```
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

### 3.24 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
```

```
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep -M --\
    $'def ([^"]|"(?!""))*"""([^"]|"(?!""))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_empty_line_after_python_function_docstring\
    "${LFBFL_common_options[@]}"
}
```

### 3.25 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
```

```
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking import of _collections_abc is at the right place\n"

  enhanced_set_shell_option pipefail --trap-unset

  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.py" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_collections_abc_place: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      "${LFBFL_file_path}"\
      > "${LFBFL_file_path}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_path}"\
      "${LFBFL_file_path}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

```

```
  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_collections_abc_place "${LFBFL_common_options[@]}"
}
```

### 3.26 build_and_checks_dependencies/repository_data.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
```

```
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_bash_option extglob --trap-unset

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

### 3.27 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion."
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "%s\n" "${LFBFL_s}"

  pcre2grep -M -- $'\$\{#.*(\b|_)arr_(?!.*\[@\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
```

```
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_arg
  declare -i LFBFL_i_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_i_max_line_length=$((LFBFL_arg))
      break
    fi
  done
  declare -i LFBFL_i_max_previous_line_length
  LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];)"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"
  # pcre2grep -M -- '^[^\n]{1,70}(?<=\]\];)(?<!^\s{0,70}\]\];)\n\s*then'
  pcre2grep -M -- "${LFBFL_regexp}" **/*.sh
}

check_no_negation_before_bash_test(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

```

```
  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking that shell scripts do not contain ! before [[.\n"

  pcre2grep -M -- $'!\s*\[\[' **/*.sh
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose --max-line-length

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
  check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
}
```

### 3.28 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php

/**
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```

```
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".

@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once './build_and_checks_dependencies/split_score.libr.php';

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// phpcs:disable Generic.WhiteSpace.ArbitraryParenthesesSpacing
// The first cast "(int) $func..." is to avoid false
// positive errors on possible XSS if output is unescaped.
```

```
echo(
  (int) generate_split_score(
    (bool) $argv[1],
    (int) $argv[2],
    explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
// phpcs:enable Generic.WhiteSpace.ArbitraryParenthesesSpacing
?>
```

### 3.29 build_and_checks_dependencies/split_score.libr.php

```
<?php

/**
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".

@category Library
@package DevOrSysAdminScripts
```

```
35 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
36 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
37 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
38 */
39
40 declare(strict_types=1);
41 declare(encoding='UTF-8');
42
43
44
45 /**
46 Given the 3 following arguments, this function returns a closure that can
47 be used to compute the split score given the current string
48 (hopefully matching a delimiter), the current cut/split position,
49 and if that split position is after or before the current string.
50
51 @param bool  $b_larger_after
52              Score is larger when splitting after,
53              previous name of this parameter was $b_after_before.
54 @param int   $i_max_length
55              The maximum length allowed (goal length).
56 @param array $a_delimiter_strings_domain
57              An array of "delimiters".
58
59 @return Closure The "split_score" closure to use repeatedly afterward.
60 */
61 function generate_split_score(
62   bool $b_larger_after,
63   int $i_max_length,
64   array $a_delimiter_strings_domain,
65 ) : Closure {
66   return function (
67     string $s_delimiter_string,
68     int $i_cut_position,
69     bool $b_is_cut_after,
70   ) use (
71     $b_larger_after,
72     $i_max_length,
73     $a_delimiter_strings_domain,
74   ) : int {
75     if(
76       !in_array(
77         $s_delimiter_string,
78         $a_delimiter_strings_domain,
79         true,
80       )
```

```
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_larger_after){
      // When $b_larger_after,
      // always way better to cut after '/' than before it.
      // When !$b_larger_after (aka larger_before),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}//end generate_split_score()
?>
```

### 3.30 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
```

```
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

#-----------------------------------------------------------------------
#-----------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#-----------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
```

```
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
#   we can see that the score really depends on the following arguments
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
```

```
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
```

```
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 + 1 and RP7 : p2 = p1 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that’s not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
```

```
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case has no meaning:
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1). The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
```

```
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
```

```
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
#   Thus we can focus on md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
```

```
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
#   enough in my brain.
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#-----------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
```

```
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

```

```
# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
```

```
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
# at least one of the previous and next characters is a delimiter.
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

#-----------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-----------------------------------------------------------------------
#-----------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
    | SSP_DELIMITER_UNIFORM
    | SSP_EQUAL_AFTER_OR_BEFORE
    | SSP_LARGER_AFTER
    | SSP_LARGER_BEFORE
```

```
532       | SSP_POSITION_NOT_DECREASING
533       | SSP_POSITION_NOT_DECREASING_AFTER
534       | SSP_POSITION_NOT_DECREASING_BEFORE
535       | SSP_POSITION_UNIFORM
536       | SSP_POSITION_UNIFORM_AFTER
537       | SSP_POSITION_UNIFORM_BEFORE
538       | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
539       | SSP_SAME_DELIMITER_LARGER_AFTER
540       | SSP_SAME_DELIMITER_LARGER_BEFORE
541       | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
542       | SSP_SAME_POSITION_LARGER_AFTER
543       | SSP_SAME_POSITION_LARGER_BEFORE
544       | SSP_UNIFORM
545     ))
546     until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
547       LFBFL_i_result_old=$((LFBFL_i_result))
548       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
549         LFBFL_i_result=$((
550           LFBFL_i_result
551           | SSP_LARGER_AFTER
552           | SSP_SAME_DELIMITER_LARGER_AFTER
553           | SSP_SAME_POSITION_LARGER_AFTER
554         ))
555       fi
556       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
557         LFBFL_i_result=$((
558           LFBFL_i_result
559           | SSP_LARGER_BEFORE
560           | SSP_SAME_DELIMITER_LARGER_BEFORE
561           | SSP_SAME_POSITION_LARGER_BEFORE
562         ))
563       fi
564       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
565         LFBFL_i_result=$((
566           LFBFL_i_result
567           | SSP_POSITION_NOT_DECREASING_AFTER
568           | SSP_POSITION_NOT_DECREASING_BEFORE
569         ))
570       fi
571       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
572         LFBFL_i_result=$((
573           LFBFL_i_result
574           | SSP_POSITION_UNIFORM_AFTER
575           | SSP_POSITION_UNIFORM_BEFORE
576           | SSP_POSITION_NOT_DECREASING
577           | SSP_POSITION_NOT_DECREASING_AFTER
```

```
578         | SSP_POSITION_NOT_DECREASING_BEFORE
579       ))
580     fi
581     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
582       LFBFL_i_result=$((
583         LFBFL_i_result
584         | SSP_POSITION_NOT_DECREASING_AFTER
585       ))
586     fi
587     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
588       LFBFL_i_result=$((
589         LFBFL_i_result
590         | SSP_POSITION_NOT_DECREASING_BEFORE
591       ))
592     fi
593     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
594       LFBFL_i_result=$((
595         LFBFL_i_result
596         | SSP_EQUAL_AFTER_OR_BEFORE
597         | SSP_LARGER_AFTER
598         | SSP_LARGER_BEFORE
599         | SSP_SAME_DELIMITER_LARGER_AFTER
600         | SSP_SAME_DELIMITER_LARGER_BEFORE
601       ))
602     fi
603     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
604       LFBFL_i_result=$((
605         LFBFL_i_result
606         | SSP_LARGER_AFTER
607       ))
608     fi
609     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
610       LFBFL_i_result=$((
611         LFBFL_i_result
612         | SSP_LARGER_BEFORE
613       ))
614     fi
615     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
616       LFBFL_i_result=$((
617         LFBFL_i_result
618         | SSP_DELIMITER_UNIFORM
619         | SSP_EQUAL_AFTER_OR_BEFORE
620         | SSP_LARGER_AFTER
621         | SSP_LARGER_BEFORE
622         | SSP_SAME_POSITION_LARGER_AFTER
623         | SSP_SAME_POSITION_LARGER_BEFORE
```

```
624         ))
625       fi
626       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
627         LFBFL_i_result=$((
628           LFBFL_i_result
629           | SSP_LARGER_AFTER
630         ))
631       fi
632       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
633         LFBFL_i_result=$((
634           LFBFL_i_result
635           | SSP_LARGER_BEFORE
636         ))
637       fi
638       if ((LFBFL_i_result & SSP_UNIFORM)); then
639         LFBFL_i_result=$((LFBFL_i_all_flags))
640         break
641       fi
642       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
643       && ((
644         LFBFL_i_result & (
645           SSP_ALWAYS_LARGER_BEFORE
646           | SSP_EQUAL_AFTER_OR_BEFORE
647           | SSP_LARGER_BEFORE
648           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
649           | SSP_SAME_DELIMITER_LARGER_BEFORE
650           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
651           | SSP_SAME_POSITION_LARGER_BEFORE
652         )
653       )); then
654         LFBFL_i_result=$((LFBFL_i_all_flags))
655         break
656       fi
657       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
658       && ((
659         LFBFL_i_result & (
660           SSP_EQUAL_AFTER_OR_BEFORE
661           | SSP_LARGER_AFTER
662           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
663           | SSP_SAME_DELIMITER_LARGER_AFTER
664           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
665           | SSP_SAME_POSITION_LARGER_AFTER
666         )
667       )); then
668         LFBFL_i_result=$((LFBFL_i_all_flags))
669         break
```

```
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_LARGER_AFTER))\
    && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_EQUAL_AFTER_OR_BEFORE
```

```
716         ))
717       fi
718       if ((LFBFL_i_result & SSP_LARGER_AFTER))\
719       && ((
720         LFBFL_i_result & (
721           SSP_POSITION_UNIFORM
722           | SSP_POSITION_UNIFORM_AFTER
723         )
724       )); then
725         LFBFL_i_result=$((
726           LFBFL_i_result
727           | SSP_SAME_DELIMITER_LARGER_AFTER
728         ))
729       fi
730       if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
731       && ((
732         LFBFL_i_result & (
733           SSP_POSITION_UNIFORM
734           | SSP_POSITION_UNIFORM_BEFORE
735         )
736       )); then
737         LFBFL_i_result=$((
738           LFBFL_i_result
739           | SSP_SAME_DELIMITER_LARGER_BEFORE
740         ))
741       fi
742       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
743       && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
744         LFBFL_i_result=$((
745           LFBFL_i_result
746           | SSP_POSITION_NOT_DECREASING
747         ))
748       fi
749       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
750       && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
751         LFBFL_i_result=$((
752           LFBFL_i_result
753           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
754         ))
755       fi
756       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
757       && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
758         LFBFL_i_result=$((LFBFL_i_all_flags))
759         break
760       fi
761       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
```

```
762     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
763       LFBFL_i_result=$((
764         LFBFL_i_result
765         | SSP_ALWAYS_LARGER_AFTER
766       ))
767     fi
768     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
769     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
770       LFBFL_i_result=$((
771         LFBFL_i_result
772         | SSP_ALWAYS_LARGER_BEFORE
773       ))
774     fi
775     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
776     && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
777       LFBFL_i_result=$((
778         LFBFL_i_result
779         | SSP_POSITION_UNIFORM
780       ))
781     fi
782     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
783     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
784       LFBFL_i_result=$((
785         LFBFL_i_result
786         | SSP_ALWAYS_LARGER_AFTER
787       ))
788     fi
789     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
790     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
791       LFBFL_i_result=$((
792         LFBFL_i_result
793         | SSP_ALWAYS_LARGER_BEFORE
794       ))
795     fi
796     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
797     && ((
798       LFBFL_i_result & (
799         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
800         | SSP_SAME_POSITION_LARGER_AFTER
801         | SSP_SAME_POSITION_LARGER_BEFORE
802       )
803     )); then
804       LFBFL_i_result=$((LFBFL_i_all_flags))
805       break
806     fi
807     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
```

```
808       && ((
809         LFBFL_i_result & (
810           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
811           | SSP_SAME_POSITION_LARGER_BEFORE
812         )
813       )); then
814         LFBFL_i_result=$((LFBFL_i_all_flags))
815         break
816       fi
817       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
818       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
819         LFBFL_i_result=$((
820           LFBFL_i_result
821           | SSP_ALWAYS_LARGER_AFTER
822         ))
823       fi
824       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
825       && ((
826         LFBFL_i_result & (
827           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
828           |  SSP_SAME_POSITION_LARGER_AFTER
829         )
830       )); then
831         LFBFL_i_result=$((LFBFL_i_all_flags))
832         break
833       fi
834       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
835       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
836         LFBFL_i_result=$((
837           LFBFL_i_result
838           | SSP_ALWAYS_LARGER_BEFORE
839         ))
840       fi
841       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
842       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
843         LFBFL_i_result=$((
844           LFBFL_i_result
845           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
846         ))
847       fi
848     done
849     declare -gi i_split_score_properties_logical_closure_result=$((
850       LFBFL_i_result
851     ))
852   }
853
```

```
854 split_score_simple(){
855   # $1=after_before
856   # $2=delimiters_strings_domain concatenated characters/delimiters
857   # $3=delimiter_string a single character
858   # $4=is_cut_after
859   declare -gi i_split_score_result
860   if [[ "$2" != *$3* ]]; then
861     i_split_score_result=0
862     return
863   fi
864   if [[ "$4" == "$1" ]]; then
865     i_split_score_result=2
866     return
867   fi
868   i_split_score_result=1
869 }
870
871 get_split_score_simple(){
872   # $1=after_before
873   # $2=max_length
874   # $3=delimiters_strings_domain
875   declare -g get_split_score_result="split_score_simple $1 '$3'"
876   declare -gi i_get_split_score_result2
877   # i_get_split_score_result2=$((11-$1*4))
878   # if [[ $1 -eq 1 ]]; then
879   #   i_get_split_score_result2=7
880   # else
881   #   i_get_split_score_result2=11
882   # fi
883   i_get_split_score_result2=$((
884     SSP_DELIMITER_UNIFORM
885     + SSP_POSITION_UNIFORM
886     + SSP_LARGER_BEFORE
887     + (
888       $1 * (
889         SSP_LARGER_AFTER
890         - SSP_LARGER_BEFORE
891       )
892     )
893   ))
894   # shellcheck disable=SC2248
895   split_score_properties_logical_closure ${i_get_split_score_result2}
896   i_get_split_score_result2=$((
897     i_split_score_properties_logical_closure_result
898   ))
899 }
```

```
900
901 get_split_score_exec(){
902   declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
903   SPLIT_SCORE_EXEC+="/split_score.exec.php"
904 }
905
906 split_score(){
907   # $1=after_before
908   # $2=max_length
909   # $3=delimiters_strings_domain concatenated characters/delimiters
910   # $4=delimiter_string a single character
911   # $5=cut_position
912   # $6=is_cut_after
913   get_split_score_exec
914   declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
915   declare -gi i_split_score_result
916   i_split_score_result=$(eval "${LFBFL_command}")
917 }
918
919 # This function is currently not called,
920 # hence we need this shellcheck disable.
921 # shellcheck disable=SC2034
922 get_split_score(){
923   # $1=after_before
924   # $1=max_length
925   # $2=delimiters_strings_domain
926   declare -g get_split_score_result="split_score $1 $2 '$3'"
927   declare -gi i_get_split_score_result2
928   # i_get_split_score_result2=$((9-$1*4))
929   i_get_split_score_result2=$((
930     SSP_DELIMITER_UNIFORM
931     + SSP_LARGER_BEFORE
932     + (
933       $1 * (
934         SSP_LARGER_AFTER
935         - SSP_LARGER_BEFORE
936       )
937     )
938   ))
939   # shellcheck disable=SC2248
940   split_score_properties_logical_closure ${i_get_split_score_result2}
941   i_get_split_score_result2=$((
942     i_split_score_properties_logical_closure_result
943   ))
944 }
945
```

```
split_line_at_most(){
  # $1=line
  # $2=max_length of beginning result string
  # $3=split_score_command
  #   negative score means do not consider as a possible split
  # $4=split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   Old values to get a simple idea without reading everything at the
  #   beginning of file:
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=forbidden_previous_character
  # $6=split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
  #     in that case, only mono-character strings are considered,
  #   - this argument can be an array of characters,
  #     optimize accordingly,
  #   - this argument can be an array of strings,
  #     optimize accordingly,
  #   - this argument can be an array of regexps,
  #     optimize accordingly.
  #     Not sure an array of regexps makes more sense than a single
  #     one. Maybe they could be both cases with corresponding
  #     optimizations. Same for a single character or substring.
  #
  # Note that we're always looking for the last position to split
  # among the positions with maximum score;
  # if some position gets a score from 2 delimiters,
  # its score is always the maximum of the two corresponding scores.
  declare -g split_line_at_most_result_start
  declare -g split_line_at_most_result_end
  declare -A LFBFL_positions
  # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
  # For my use case in Bash scripts, I will need only an array of
  # characters. See get_split_score().
  if [[ -n "$6" ]]; then
    printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
  fi
```

```
992     if [[ $2 -ge ${#1} ]]; then
993       split_line_at_most_result_start=$1
994       split_line_at_most_result_end=""
995       return
996     fi
997     # We know that ${#1} > $2, hence ${#1} - 1 >= $2
998     # Edge cases:
999     # There is always at least one character that is overlength,
1000    # but we need to look only at the score before that character that is
1001    # overlength, the score after that character is invalid,
1002    # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
1003    # That first character that is overlength has index $2.
1004    # The cut position after that character has index $2 + 1.
1005    # Looping from character position going from 0 to $2, we need to drop
1006    # score before character 0 and drop score after character $2.
1007    # We do this with forbidden score -1.
1008    declare -ir LFBFL_i_max=$(($2))
1009    declare -i LFBFL_i
1010    declare -i LFBFL_j
1011    local LFBFL_current_char
1012
1013    # At the beginning, each position has score 0.
1014    LFBFL_positions["0"]="-1"
1015    for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1016      LFBFL_positions["${LFBFL_i}"]="0"
1017    done
1018    LFBFL_positions["${LFBFL_i}"]="-1"
1019
1020    # If we have a forbidden previous character,
1021    # we mark the forbidden positions with -1.
1022    if [[ -n "$5" ]]; then
1023      for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1024        LFBFL_current_char="${1:${LFBFL_i}:1}"
1025        ((++LFBFL_i))
1026        if [[ "${LFBFL_current_char}" == "$5" ]]; then
1027          LFBFL_positions["${LFBFL_i}"]="-1"
1028        fi
1029      done
1030    fi
1031
1032    declare -r LFBFL_sort_command="sort --numeric-sort"
1033
1034    if (($4 & SSP_DELIMITER_UNIFORM))\
1035    && (($4 & SSP_LARGER_AFTER))\
1036    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1037      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
```

```
1038        LFBFL_current_char="${1:${LFBFL_i}:1}"
1039        eval "$3 '${LFBFL_current_char}' 1"
1040        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1041        if [[ i_split_score_result -ge 1 ]]; then
1042          LFBFL_j=$((LFBFL_i + 1))
1043          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1044          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1045            LFBFL_positions["${LFBFL_j}"]=$(
1046              max "${LFBFL_sort_command}"\
1047                "${LFBFL_positions["${LFBFL_j}"]}"\
1048                "${i_split_score_result}"
1049            )
1050            break
1051          fi
1052        fi
1053        eval "$3 '${LFBFL_current_char}' 0"
1054        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1055        if [[ i_split_score_result -ge 1 ]]; then
1056          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1057          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1058            LFBFL_positions["${LFBFL_i}"]=$(
1059              max "${LFBFL_sort_command}"\
1060                "${LFBFL_positions["${LFBFL_i}"]}"\
1061                "${i_split_score_result}"
1062            )
1063          fi
1064        fi
1065      done
1066    elif (($4 & SSP_DELIMITER_UNIFORM))\
1067    && (($4 & SSP_LARGER_BEFORE))\
1068    && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
1069      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1070        LFBFL_current_char="${1:${LFBFL_i}:1}"
1071        # We still need to break in second if, after computing score after,
1072        # because SSP_LARGER_BEFORE implies large inequality instead of
1073        # strict inequality.
1074        # And since the algorithm is to take the largest position with
1075        # maximum score, there is a possibility that we'll break on a found
1076        # score before but for splitting on a score after.
1077        eval "$3 '${LFBFL_current_char}' 1"
1078        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1079        if [[ i_split_score_result -ge 1 ]]; then
1080          LFBFL_j=$((LFBFL_i + 1))
1081          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1082          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1083            LFBFL_positions["${LFBFL_j}"]=$(
```

```
1084            max "${LFBFL_sort_command}"\
1085              "${LFBFL_positions["${LFBFL_j}"]}"\
1086              "${i_split_score_result}"
1087          )
1088        fi
1089      fi
1090      eval "$3 '${LFBFL_current_char}' 0"
1091      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1092      if [[ i_split_score_result -ge 1 ]]; then
1093        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1094        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1095          LFBFL_positions["${LFBFL_i}"]=$(
1096            max "${LFBFL_sort_command}"\
1097              "${LFBFL_positions["${LFBFL_i}"]}"\
1098              "${i_split_score_result}"
1099          )
1100          break
1101        fi
1102      fi
1103    done
1104  else
1105    for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
1106      LFBFL_current_char="${1:${LFBFL_i}:1}"
1107      eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1108      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1109      if [[ i_split_score_result -ge 1 ]]; then
1110        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1111        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1112          LFBFL_positions["${LFBFL_i}"]=$(
1113            max "${LFBFL_sort_command}"\
1114              "${LFBFL_positions["${LFBFL_i}"]}"\
1115              "${i_split_score_result}"
1116          )
1117        fi
1118      fi
1119      ((++LFBFL_i))
1120      eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
1121      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1122      if [[ i_split_score_result -ge 1 ]]; then
1123        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1124        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1125          LFBFL_positions["${LFBFL_i}"]=$(
1126            max "${LFBFL_sort_command}"\
1127              "${LFBFL_positions["${LFBFL_i}"]}"\
1128              "${i_split_score_result}"
1129          )
```

```
        fi
      fi
    done
  fi
  declare -ir LFBFL_i_max_score=$(
    max 'sort --numeric-sort' "${LFBFL_positions[@]}"
  )
  local LFBFL_value
  declare -i LFBFL_i_best_position
  for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
    if [[ LFBFL_i -eq 0 ]]; then
      printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
    fi
    local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
    # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
      LFBFL_i_best_position=$((LFBFL_i))
      break
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_i_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=new_lines
  # $2=prefix
  # $3=max_length
  # $4=suffix
  # $5=split_score_command
  # $6=split_score_command_properties
  # $7=forbidden_previous_character ('\' usually)
  # Options:
  #   --verbose
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
```

```
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_i_overlength=$(($3 + 1))
  declare -ir LFBFL_i_length2=$(($3 - ${#4}))
  # shellcheck disable=SC2248
  get_overlength_regexp ${LFBFL_i_overlength}

  # Testing if some line is too long.
  printf "%s" "$1"\
    | grep -q "${get_overlength_regexp_result}";
  if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
    return
  fi

  declare -r LFBFL_start=$(
    printf "%s" "$1"\
    | head --lines=-1
  )
  # printf "start: %s\n" "${LFBFL_start}"
  declare -r LFBFL_last_line=$(
    printf "%s" "$1"\
    | tail --lines=1
  )
  # printf "last_line: %s\n" "${LFBFL_last_line}"
  split_last_line_result=""
  if [[ -n "${LFBFL_start}" ]]; then
```

```
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
  fi
  if [[ -n "$5" ]]; then
    # shellcheck disable=SC2248
    split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
```

```
    fi
  fi
}
```

### 3.31 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  declare -g get_overlength_regexp_result=".\{$1\}"
  # declare -g get_overlength_regexp_result="^.\{$1\}"
  # declare -g get_overlength_regexp_result=".\{$1\}\$"
  # First one seems to be faster by a few percents when testing like this:
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
```

```
  # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
}

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_arg
  declare -i LFBFL_i_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_i_max_line_length=$((LFBFL_arg))
      break
    fi
  done

  declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
  # shellcheck disable=SC2248
  get_overlength_regexp ${LFBFL_i_overlength}

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  local LFBFL_long_line
  local LFBFL_file_path
  local LFBFL_line
  local LFBFL_extension
  local LFBFL_base_name
  local LFBFL_s_long_lines
  declare -a LFBFL_arr_long_lines
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
      printf "too_long_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    # shellcheck disable=SC2248
    LFBFL_s_long_lines=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | xargs grep --line-number --with-filename\
        "${get_overlength_regexp_result}"
    )
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "too_long_code_lines: No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      # Remove file_path and line_number
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.sub"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

### 3.32 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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# or (at your option) any later version.
#
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# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp "./${LFBFL_common_file_name}.tpl"\
```

```
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_files_names=(
    "build_md_from_printable_md.exec.sh"
    "build_dependencies_notes.exec.php"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spelling_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
    "overwrite_if_not_equal.libr.sh"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "repository_data.libr.sh"
    "shell_checks_complement.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut -f1 -d' '
    )
    LFBFL_file_sha512_all="LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
```

```
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

### 3.33 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
```

```
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

```

```
  declare -i LFBFL_i_write=0
  if [[ "$*" == *--write* ]]; then
    LFBFL_i_write=1
  fi
  readonly LFBFL_i_write

  declare -i LFBFL_i_append=0
  if [[ "$*" == *--append* ]]; then
    LFBFL_i_append=1
  fi
  readonly LFBFL_i_append

  enhanced_set_shell_option pipefail --trap-unset

  LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
  readonly LFBFL_listing
  local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
  readonly LFBFL_data_file_name

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_i_"

  if [[ LFBFL_i_write -eq 1 ]]; then
    : > "${LFBFL_listing}"
  fi
  # Remove line returns that are here to keep lines short.
  sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
    > "${LFBFL_listing}.temp"
  # shopt -s dotglob was needed at some point but I don't see why now.
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} /
  declare -r LFBFL_split_score_command="${get_split_score_result}"
  declare -i LFBFL_i_split_score_command_properties
  LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
  readonly LFBFL_i_split_score_command_properties
  local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
  readonly LFBFL_suffix
  # There is always a $'\n' final suffix from >>.
  # But when origin string ends with the suffix we must add something.
  declare -r LFBFL_final_suffix=$'\n\\'

  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find . -type f -printf '%P\n'\
```

```
    | relevant_find\
    | sort
  )
  # printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf\
        "update_or_check_files_names_listing: Some files to process.\n"
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  # printf "%s\n" "${LFBFL_arr_files_paths[@]}"

  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    # if [[ LFBFL_i_verbose -eq 1 ]]; then
    #   printf\
    #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
    #     "${LFBFL_file_path}"
    # fi
    git check-ignore -q "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
        "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_i_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      ${LFBFL_i_split_score_command_properties}\
      ""\
      "${LFBFL_final_suffix}"
    if [[ LFBFL_i_write -eq 1 ]]; then
      # shellcheck disable=SC2001
```

```
148         printf "%s\n" "${repeated_split_last_line_result}"\
149           >> "${LFBFL_listing}"
150         continue
151       fi
152       if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
153           "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
154       then
155         :
156       else
157         printf "The file %s is not listed in %s.\n"\
158           "${LFBFL_file_path}"\
159           "${LFBFL_listing}"
160         if [[ LFBFL_i_append -eq 1 ]]; then
161           # shellcheck disable=SC2001
162           printf "%s\n" "${repeated_split_last_line_result}"\
163             >> "${LFBFL_listing}"
164         fi
165       fi
166     done
167
168     LFBFL_s_files_paths=$(cat "${LFBFL_listing}.temp")
169     if [[ -n "${LFBFL_s_files_paths}" ]]; then
170       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
171     fi
172
173     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
174       [[ "${LFBFL_file_path}" != '//'* ]] || continue
175       if [[ ! -f "${LFBFL_file_path}" ]]; then
176         printf "The non-file %s is listed in %s.\n"\
177           "${LFBFL_file_path}"\
178           "${LFBFL_listing}"
179       fi
180     done
181
182     rm "${LFBFL_listing}.temp"
183
184     if [[ LFBFL_i_verbose -eq 1 ]]; then
185       cat "${LFBFL_listing}"
186     fi
187   }
188
189   update_or_check_files_names_listing "$@"
```

### 3.34 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts
```

### 3.35 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

### 3.36 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

### 3.37 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

### 3.38 build_and_checks_variables/dependencies_data.json

```
[
  { "dependency_name": "Bash", "URLs": [
      "https://www.gnu.org/software/bash/",
      "https://tiswww.case.edu/php/chet/bash/bashtop.html",
      "https://savannah.gnu.org/projects/bash/"
  ] },
  { "dependency_name": "GNU coreutils", "URLs": [
      "https://www.gnu.org/software/coreutils/",
      "https://savannah.gnu.org/projects/coreutils/"
  ] },
  { "dependency_name": "uutils coreutils", "URLs": [
      "https://uutils.github.io/coreutils/",
      "https://github.com/uutils/coreutils"
  ] },
  { "dependency_name": "diff (GNU diffutils)", "URLs": [
      "https://www.gnu.org/software/diffutils/",
      "https://savannah.gnu.org/projects/diffutils/"
  ] },
  { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
      "https://www.gnu.org/software/findutils/",
      "https://savannah.gnu.org/projects/findutils/"
  ] },
  { "dependency_name": "grep (GNU grep)", "URLs": [
      "https://www.gnu.org/software/grep/",
      "https://savannah.gnu.org/projects/grep/"
```

```
26      ] },
27      { "dependency_name": "sed (GNU sed)", "URLs": [
28          "https://www.gnu.org/software/sed/",
29          "https://savannah.gnu.org/projects/sed/"
30      ] },
31      { "dependency_name": "ShellCheck", "URLs": [
32          "https://www.shellcheck.net/",
33          "https://github.com/koalaman/shellcheck"
34      ] },
35      { "dependency_name": "GNU Wget", "URLs": [
36          "https://www.gnu.org/software/wget/",
37          "https://savannah.gnu.org/projects/wget/"
38      ] },
39      { "dependency_name": "more of util-linux", "URLs": [
40          "https://www.kernel.org/pub/linux/utils/util-linux/",
41          "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42          {
43            "URL": "https://github.com/util-linux/util-linux",
44            "comment": "official?"
45          }
46      ] },
47      { "dependency_name": "pandoc", "URLs": [
48          "https://pandoc.org/",
49          "https://github.com/jgm/pandoc"
50      ] },
51      { "dependency_name": "pdftex/pdflatex", "URLs": [
52          "https://www.tug.org/applications/pdftex/",
53          "https://tug.org/texlive/svn/"
54      ] },
55      { "dependency_name": "tree", "URLs": [
56          "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57          "https://gitlab.com/OldManProgrammer/unix-tree",
58          "https://github.com/Old-Man-Programmer/tree"
59      ] },
60      { "dependency_name": "git", "URLs": [
61          "https://git-scm.com/about",
62          "https://github.com/git/git"
63      ] },
64      { "dependency_name": "pylint", "URLs": [
65          "https://www.pylint.org/",
66          "https://pylint.readthedocs.io/en/stable/",
67          "https://github.com/pylint-dev/pylint",
68          "https://pypi.org/project/pylint/"
69      ] },
70      { "dependency_name": "notify-send", "URLs": [
71          "https://gitlab.gnome.org/GNOME/libnotify/",
```

```
72          "https://gnome.pages.gitlab.gnome.org/libnotify/"
73      ] },
74      { "dependency_name": "black", "URLs": [
75          "https://github.com/psf/black",
76          "https://black.readthedocs.io/en/stable/",
77          "https://pypi.org/project/black/"
78      ] },
79      { "dependency_name": "isort", "URLs": [
80          "https://pycqa.github.io/isort/",
81          "https://github.com/pycqa/isort/",
82          "https://pypi.org/project/isort/"
83      ] },
84      { "dependency_name": "mypy", "URLs": [
85          "https://mypy-lang.org/",
86          "https://mypy.readthedocs.io/en/stable/",
87          "https://github.com/python/mypy",
88          "https://pypi.org/project/mypy/"
89      ] },
90      { "dependency_name": "pip", "URLs": [
91          "https://pip.pypa.io/en/stable/",
92          "https://github.com/pypa/pip",
93          {
94            "URL": "https://pypi.org/project/pip/",
95            "comment": "(egg and chicken ;) )"
96          }
97      ] },
98      { "dependency_name": "ruff", "URLs": [
99          "https://docs.astral.sh/ruff/",
100         "https://github.com/astral-sh/ruff",
101         "https://pypi.org/project/ruff/"
102     ] },
103     { "dependency_name": "bandit", "URLs": [
104         "https://bandit.readthedocs.io/en/latest/",
105         "https://github.com/PyCQA/bandit",
106         "https://pypi.org/project/bandit/"
107     ] },
108     { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
109         "https://github.com/PHPCSStandards/PHP_CodeSniffer",
110         "https://packagist.org/packages/squizlabs/php_codesniffer"
111     ] },
112     { "dependency_name": "phpmd", "URLs": [
113         "https://phpmd.org/",
114         "https://github.com/phpmd/phpmd",
115         "https://packagist.org/packages/phpmd/phpmd"
116     ] },
117     { "dependency_name": "php", "URLs": [
```

```
        "https://www.php.net/",
        "https://github.com/php/php-src"
    ] },
    { "dependency_name": "tex", "URLs": [
        "https://ctan.org/tex-archive/systems/knuth/dist/tex"
    ] },
    { "dependency_name": "pcregrep, pcre2grep", "URLs": [
        "https://www.pcre.org/",
        "https://pcre2project.github.io/pcre2/",
        "https://github.com/PCRE2Project/pcre1",
        "https://github.com/PCRE2Project/pcre2",
        "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
    ] },
    { "dependency_name": "python3", "URLs": [
        "https://www.python.org/",
        "https://docs.python.org/3/"
    ] }
]
```

### 3.39 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

```

```
sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
```

```
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer
https://packagist.org/packages/squizlabs/php_codesniffer

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
```

```
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

### 3.40 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
```

```
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have an accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
```

```
78 ol.level2{
79   counter-reset: level2;
80 }
81 ol.level1 li:before{
82   content: counter(level1) ". ";
83   counter-increment: level1;
84 }
85 ol.level2 li:before{
86   content: counter(level1) "." counter(level2) " ";
87   counter-increment: level2;
88 }
89 </style>
90 <title>@repository_name@</title>
91 </head>
92 <body >
93 <table class="title">
94   <tr>
95     <td style="padding:1ex">
96       <h1 class="titlemain">@repository_name@</h1>
97       <h2 class="titlerest">
98         @author_full_name@
99         <br>
100        <a href="@author_website@">
101          <span style="font-family:monospace">
102          @author_website@
103          </span>
104        </a>
105        <br>
106        <a href="mailto:@author_email@">
107          <span style="font-family:monospace">
108          @author_email@
109          </span>
110        </a>
111      </h2>
112    </td>
113  </tr>
114 </table>
115
116 <blockquote class="abstract">
117   <span style="font-weight:bold">Abstract:</span>
118   @abstract@
119 </blockquote>
120 <p>Current version: @current_date@</p>
121 <p>Current number of commits: @number_of_commits@</p>
122 <p>Current git SHA1: @current_git_SHA1@</p>
123 <p>
```

```
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Notes on licenses and dependencies</a></li>
  <li><a href="#section3">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Notes on licenses and dependencies</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
```

```
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpcs, phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@
```

```

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

### 3.41 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
```

```
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
```

```
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

```

```
Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpcs, phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

\section{Listing of files}
\label{section:listing}

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

### 3.42 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
```

```
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_dependencies_notes.libr.php
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
```

```
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpcs_ruleset.xml
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
```

```
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234567890123456\
56
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234567890123456\
56\
\
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234567890123456\
5\
\
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234567890123456\
\
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234567890123\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

### 3.43 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
```

### 3.44 build_and_checks_variables/phpcs_ruleset.xml

```
<?xml version="1.0"?>
<ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
  <description>A custom coding standard</description>
  <config name="tab_width" value="2"/>

  <file>../</file>

  <arg name="extensions" value="php"/>
```

```
  <arg name="report" value="summary"/>
  <arg name="colors"/>
  <arg value="sp"/>

  <!--
  I added all the standards, then excluded all the rules that gave me
  false positives or that seemed unappropriate when looking at the
  documentation.
  Some are bugs, some are conflicting with my own coding style.
  You may delete an exclude and see what pops up to maybe understand
  my choices.
  -->

  <rule ref="Generic">
    <!-- You have to allow long or short or both syntaxes. -->
    <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
    <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
    <!--
    Not standard in Unix/Linux world to keep files not ending with \n.
    This is always an option of the code editors.
    Since PHP doesn't output the final \n after ?>, it is ok.
    -->
    <exclude name="Generic.Files.EndFileNoNewline.Found"/>
    <exclude name="Generic.Files.ExecutableFile.Executable"/>
    <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
    <exclude
     name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
    />
    <exclude
      name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
    />
    <!-- Applies only to true, false and null. -->
    <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
    <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
    <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
    <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
  </rule>
  <rule ref="PEAR">
    <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
    <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
    <!--
    Look at the comment before generate_split_score.
    PEAR and Squiz require to align the parameter names parts in the same
    comment, and also the parameter comments parts in the same comment,
```

```
55      but with a long parameter name like $a_delimiter_strings_domain
56      and a line length limit at 75 characters, parameter comments parts
57      have only 24 characters left per line, which is ridiculous.
58      Our choice to put the parameter comments parts starting on another line
59      with same "indent" as parameter names parts is way better in those
60      cases. Alignment has always been a trap with short lines.
61      Since phpDocumentor handles the parameter comments parts correctly,
62      it is enough for me.
63      -->
64      <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
65      <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
66      <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
67      <exclude
68        name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
69      />
70      <exclude
71       name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
72      />
73      <exclude
74        name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
75      />
76      <!--
77      Don't laugh when looking at the false positives of the two exclude
78      below ;) XD.
79      They don't check for multiline function declarations apparently.
80      -->
81      <exclude
82        name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
83      />
84      <exclude
85        name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
86      />
87      <exclude
88        name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
89      />
90      <exclude
91        name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
92      />
93    </rule>
94    <rule ref="PSR1">
95    </rule>
96    <rule ref="PSR2">
97      <!-- Back to my habits at Teliae with always a closing tag. -->
98      <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
99      <exclude
100 name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
```

```
    />
  </rule>
  <rule ref="PSR12">
    <exclude
 name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
    />
    <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
    <!--
    3 false positives, see:
    https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
    -->
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
    />
    <!-- and even a fourth and fifth false positives -->
    <exclude
      name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
    />
    <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
    <exclude
      name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
    />
    <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
    <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
    <!-- I agree except for string concatenation. -->
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
  </rule>
  <rule ref="Squiz">
    <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
    <!--
    It doesn't know the //*/ trick...
    https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
    -->
    <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
    <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
    <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>
```

```
147     <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
148     <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
149     <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
150     <!-- Problem for doc block comment before namespace -->
151     <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
152     <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
153     <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
154     <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
155     <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
156     <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
157     <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
158     <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
159     <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
160     <!-- asks for boolean instead of bool which is incorrect -->
161     <exclude
162       name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
163     />
164     <!-- asks for boolean instead of bool which is incorrect -->
165     <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
166     <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
167     <!-- Problem for doc block comment before namespace -->
168     <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
169     <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
170     <!-- It doesn't know the //*/ trick... -->
171     <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
172     <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
173     <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
174     <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
175     <exclude
176       name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
177     />
178     <exclude
179   name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
180     />
181     <exclude
182       name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
183     />
184     <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
185     <exclude name="Squiz.Functions.GlobalFunction.Found"/>
186     <exclude
187       name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
188     />
189     <exclude
190   name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
191     />
192     <exclude
```

```
193         name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
194       />
195       <exclude
196         name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
197       />
198       <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
199       <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
200       <exclude name="Squiz.PHP.CommentedOutCode"/>
201       <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
202       <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
203       <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
204       <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>
205       <exclude
206         name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
207       />
208     </rule>
209     <rule ref="Zend">
210       <!-- Back to my habits at Teliae with always a closing tag. -->
211       <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
212       <exclude
213         name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
214       />
215     </rule>
216
217     <rule ref="Generic.Arrays.ArrayIndent">
218       <properties>
219         <property name="indent" value="2" />
220       </properties>
221     </rule>
222     <rule ref="Generic.Files.LineLength">
223       <properties>
224         <property name="lineLimit" value="75"/>
225         <property name="absoluteLineLimit" value="75"/>
226       </properties>
227     </rule>
228     <rule ref="Generic.Formatting.SpaceAfterNot">
229       <properties>
230         <property name="spacing" value="0" />
231       </properties>
232     </rule>
233     <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
234       <properties>
235         <property name="checkFunctions" value="true"/>
236         <property name="checkClosures" value="true"/>
237       </properties>
238     </rule>
```

```
239     <rule ref="Generic.Metrics.CyclomaticComplexity">
240       <properties>
241         <property name="complexity" value="20" />
242         <property name="absoluteComplexity" value="40" />
243       </properties>
244     </rule>
245     <rule ref="Generic.Strings.UnnecessaryStringConcat">
246       <properties>
247         <property name="allowMultiline" value="true" />
248       </properties>
249     </rule>
250     <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
251       <properties>
252         <property name="ignoreNewlines" value="true" />
253       </properties>
254     </rule>
255     <rule ref="Generic.WhiteSpace.ScopeIndent">
256       <properties>
257         <property name="ignoreIndentationTokens" type="array">
258           <element value="T_COMMENT"/>
259           <element value="T_DOC_COMMENT_OPEN_TAG"/>
260         </property>
261         <property name="indent" value="2"/>
262       </properties>
263     </rule>
264
265     <rule ref="PEAR.Functions.FunctionCallSignature">
266       <properties>
267         <property name="indent" value="2"/>
268       </properties>
269     </rule>
270     <rule ref="PEAR.Functions.FunctionDeclaration">
271       <properties>
272         <property name="indent" value="2"/>
273       </properties>
274     </rule>
275     <rule ref="PEAR.WhiteSpace.ScopeIndent">
276       <properties>
277         <property name="ignoreIndentationTokens" type="array">
278           <element value="T_COMMENT"/>
279           <element value="T_DOC_COMMENT_OPEN_TAG"/>
280         </property>
281         <property name="indent" value="2"/>
282       </properties>
283     </rule>
284
```

```
  <rule ref="PSR2.Methods.FunctionCallSignature">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>
  <rule ref="Squiz.Strings.ConcatenationSpacing">
    <properties>
      <property name="ignoreNewlines" value="true" />
    </properties>
  </rule>
  <rule ref="Squiz.WhiteSpace.FunctionSpacing">
    <properties>
      <property name="spacing" value="3" />
      <property name="spacingBeforeFirst" value="0" />
      <property name="spacingAfterLast" value="0" />
    </properties>
  </rule>
</ruleset>
```

### 3.45 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
```

```
14    />
15    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
16      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
17    />
18    <violation rule="PHPMD\Rule\CleanCode\ElseExpression"
19      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
20    />
21    <violation rule="PHPMD\Rule\Naming\LongVariable"
22      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
23    />
24    <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
25      file="build_and_checks_dependencies/split_score.libr.php"
26    />
27    <violation rule="PHPMD\Rule\Naming\LongVariable"
28      file="build_and_checks_dependencies/split_score.libr.php"
29    />
30    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
31      file="build_and_checks_dependencies/split_score.libr.php"
32    />
33    <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
34      file="string_escaping.libr.php"
35    />
36    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
37      file="string_escaping.libr.php"
38    />
39    <violation rule="PHPMD\Rule\Design\LongMethod"
40      file="unicode.libr.php"
41    />
42    <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
43      file="unicode.libr.php"
44    />
45    <violation rule="PHPMD\Rule\Naming\LongVariable"
46      file="unicode.libr.php"
47    />
48    <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
49      file="unicode.libr.php"
50    />
51    <violation rule="PHPMD\Rule\CleanCode\MissingImport"
52      file="unicode.libr.php"
53    />
54    <violation rule="PHPMD\Rule\CyclomaticComplexity"
55      file="unicode.libr.php"
56    />
57    <violation rule="PHPMD\Rule\Design\NpathComplexity"
58      file="unicode.libr.php"
59    />
```

```
</phpmd-baseline>
```

### 3.46 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

### 3.47 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
```

```
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

grammar_and_spelling_checker_command=


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
```

```
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff
```

```
upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800

use_vnu_for_xhtml=false
```

### 3.48 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
```

```
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

acknowledgments=
acknowledgments=

max_line_length=

latex_command=

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=

use_vnu_for_xhtml=

npm_lint_directories=
npm_lint_directories=
```

### 3.49 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
```

```
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

### 3.50 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
```

```
source-path="./git_helpers"
```

### 3.51 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
           Use printf '..%s..' "$foo".
           False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
             # shellcheck disable=SC2145
             all_self_code_lines "$@"\
```

```
                | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162X: read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
           https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$"\
  -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$"
```

### 3.52 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
```

```
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir --parents "${LFBFL_subdir}/licenses_templates"
  local LFBFL_variables_directory
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  readonly LFBFL_variables_directory
```

```
  mkdir --parents "${LFBFL_variables_directory}/temp"
  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\nupgrade_venvs_ts\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  local LFBFL_script_download_URL
  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
  LFBFL_script_download_URL+="${LFBFL_common_file_name}"
  readonly LFBFL_script_download_URL
  declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='b46b170822e280686d3c31b028ee02991c2f32fe74164'
  LFBFL_correct_sha512+='ba80ac0702e65395be7576e06bc455f43b8e3bc8d1b6'
  LFBFL_correct_sha512+='ad585d495e1a36cf2c652f37e03f21a622e5032'
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
```

```
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

### 3.53 COPYING

```
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  The GNU General Public License does not permit incorporating your program
```

into proprietary programs. If your program is a subroutine library, you may consider it more useful to permit linking proprietary applications with the library. If this is what you want to do, use the GNU Lesser General Public License instead of this License. But first, please read <https://www.gnu.org/licenses/why-not-lgpl.html>.

### 3.54 COPYING.LESSER

```
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                       Version 3, 29 June 2007

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  An "Application" is any work that makes use of an interface provided
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Defining a subclass of a class defined by the Library is deemed a mode
of using an interface provided by the Library.

  A "Combined Work" is a work produced by combining or linking an
Application with the Library.  The particular version of the Library
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Version".

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Corresponding Source for the Combined Work, excluding any source code
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based on the Application, and not on the Linked Version.

  The "Corresponding Application Code" for a Combined Work means the
object code and/or source code for the Application, including any data
```

```
and utility programs needed for reproducing the Combined Work from the
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  If you modify a copy of the Library, and, in your modifications, a
facility refers to a function or data to be supplied by an Application
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  The object code form of an Application may incorporate material from
a header file that is part of the Library.  You may convey such object
code under terms of your choice, provided that, if the incorporated
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  You may convey a Combined Work under terms of your choice that,
taken together, effectively do not restrict modification of the
portions of the Library contained in the Combined Work and reverse
engineering for debugging such modifications, if you also do each of
```

```
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    covered by this License.

    b) Accompany the Combined Work with a copy of the GNU GPL and this license
    document.

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    execution, include the copyright notice for the Library among
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    copies of the GNU GPL and this license document.

    d) Do one of the following:

        0) Convey the Minimal Corresponding Source under the terms of this
        License, and the Corresponding Application Code in a form
        suitable for, and under terms that permit, the user to
        recombine or relink the Application with a modified version of
        the Linked Version to produce a modified Combined Work, in the
        manner specified by section 6 of the GNU GPL for conveying
        Corresponding Source.

        1) Use a suitable shared library mechanism for linking with the
        Library.  A suitable mechanism is one that (a) uses at run time
        a copy of the Library already present on the user's computer
        system, and (b) will operate properly with a modified version
        of the Library that is interface-compatible with the Linked
        Version.

    e) Provide Installation Information, but only if you would otherwise
    be required to provide such information under section 6 of the
    GNU GPL, and only to the extent that such information is
    necessary to install and execute a modified version of the
    Combined Work produced by recombining or relinking the
    Application with a modified version of the Linked Version. (If
    you use option 4d0, the Installation Information must accompany
    the Minimal Corresponding Source and Corresponding Application
    Code. If you use option 4d1, you must provide the Installation
    Information in the manner specified by section 6 of the GNU GPL
    for conveying Corresponding Source.)

  5. Combined Libraries.

  You may place library facilities that are a work based on the
```

```
Library side by side in a single library together with other library
facilities that are not Applications and are not covered by this
License, and convey such a combined library under terms of your
choice, if you do both of the following:

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   on the Library, uncombined with any other library facilities,
   conveyed under the terms of this License.

   b) Give prominent notice with the combined library that part of it
   is a work based on the Library, and explaining where to find the
   accompanying uncombined form of the same work.

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permanent authorization for you to choose that version for the
Library.
```

### 3.55 git_helpers/git_checks.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  declare -i LFBFL_i_error=0
  git diff --cached --name-only\
```

```
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_i_error=1
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_no_abusive_trailing_comma(){
  # Useful for Python code, although it regularly yields
  # false positives in Python code
  # (but it is easy to distinguish by hand).
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION"
  local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
  LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
  readonly LFBFL_send_body_1
  declare -i LFBFL_i_error=0
  git diff --cached -r\
    > /tmp/DOSAS_django_git_check2.temp
  if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_s_files_paths=$(
    git diff --cached --name-only\
    | grep '\.py'
```

```
  )
  printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  declare -i LFBFL_i_error=0
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file_path
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file_path}"
    if ! black --check --diff "${LFBFL_file_path}";
    then
      LFBFL_i_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

### 3.56 git_helpers/post-commit

```
#!/usr/bin/env bash
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# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
```

```
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

### 3.57 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
#
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# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
```

```
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_arg
  local LFBFL_s_format

  declare -i LFBFL_i_log_directory_change=0
  if [[ "$*" == *--log-directory-change* ]]; then
    LFBFL_i_log_directory_change=1
  fi
  readonly LFBFL_i_log_directory_change

  declare -i LFBFL_i_max_comment_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
      LFBFL_i_max_comment_line_length=$((LFBFL_arg))
      break
    fi
  done
  readonly LFBFL_i_max_comment_line_length

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
```

```
    # shellcheck disable=SC2059,SC2248
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  #-----------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log -p -1\
    | grep '^diff --git' -A 1
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi

  declare -a LFBFL_renaming_lines_array
  mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
  readonly LFBFL_renaming_lines_array

  declare -r LFBFL_timestamp=$(date --iso-8601="minute")
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
  fi
  local LFBFL_split_score_command1
  declare -i LFBFL_i_split_score_command_properties1
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
  LFBFL_split_score_command1="${get_split_score_result}"
  LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
  readonly LFBFL_split_score_command1
  readonly LFBFL_i_split_score_command_properties1
  local LFBFL_split_score_command2
  declare -i LFBFL_i_split_score_command_properties2
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
  LFBFL_split_score_command2="${get_split_score_result}"
  LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
  readonly LFBFL_split_score_command2
  readonly LFBFL_i_split_score_command_properties2
  declare -i LFBFL_i_renaming_happened=0
  declare -i LFBFL_i
  declare -i LFBFL_j
  # declare -r LFBFL_special_file_name="files_names_listing.txt"
```

```
141   declare -r LFBFL_sfn="files_names_listing.txt"
142   local LFBFL_diff_line
143   local LFBFL_similarity_line
144   local LFBFL_old_file_path
145   local LFBFL_old_file_name
146   local LFBFL_old_file_directory
147   local LFBFL_new_file_path
148   local LFBFL_new_file_name
149   local LFBFL_new_file_directory
150   local LFBFL_useful_file_name
151   local LFBFL_extension
152   local LFBFL_comment_prefix
153   local LFBFL_comment_prefix2
154   local LFBFL_old_file_name2
155   local LFBFL_old_file_directory2
156   local LFBFL_new_file_name2
157   local LFBFL_new_file_directory2
158   local LFBFL_new_comment
159   declare -i LFBFL_i_copyright_line_number
160   declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
161   declare -i LFBFL_i_line_count
162   declare -i LFBFL_i_lines_after
163   local LFBFL_temp_file_path
164   for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
165     LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
166     LFBFL_i=$((LFBFL_i + 1))
167     LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
168     LFBFL_i=$((LFBFL_i + 1))
169     # We need to increment 3 times since grep adds "--" lines between.
170     LFBFL_i=$((LFBFL_i + 1))
171     if [[ LFBFL_i_verbose -eq 1 ]]; then
172       printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
173         "${LFBFL_diff_line}"\
174         "${LFBFL_similarity_line}"
175     fi
176     if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
177       continue
178     fi
179
180     # Extract old file name from diff line. --------------------------
181     LFBFL_old_file_path=$(
182       printf "%s" "${LFBFL_diff_line}"\
183       | cut -d ' ' -f 3
184     )
185     # It starts by "a/".
186     LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
```

```
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. -----------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
```

```
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
```

```
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
    ]];
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
```

```
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_i_copyright_line_number=$(
      grep -n '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut -d ':' -f 1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. -------------------------------------
    LFBFL_i_line_count=$(
      wc -l\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. ---------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. ---------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
```

```
      tail --lines="${LFBFL_i_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ---------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit ---------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

### 3.58 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
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#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
```

```
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "pre-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

### 3.59 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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of the GNU Lesser General Public License
```

```
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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


```

```
def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
```

```
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
```

```
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
```

```
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
```

```
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

### 3.60 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
-----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
```

```
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

### 3.61 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
```

```
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

### 3.62 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

### 3.63 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>'
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
```

```
This is some text
</pre>

<script>
let a = 'abcd & < > \\ \n  \"\'';
alert(a);
let b = "abcd & < > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

### 3.64 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

### 3.65 README.md.tpl

```
# DevOrSysAdminScripts

[![CodeFactor-badge]][CodeFactor-package-page]
[![Codacy-badge]][Codacy-package-page]
![GitHub-top-language-badge]
![GitHub-code-size-in-bytes-badge]
[![Security:bandit:badge]][Bandit-GitHub.com]
[![Code-style:black:badge]][Black-GitHub.com]
[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
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|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
```

```
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

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llyaudet/DevOrSysAdminScripts/badge

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github/llyaudet/DevOrSysAdminScripts

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3dd6ef85e8db4460991d1ebc404b68dd

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languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

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code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black
```

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[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

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https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

### 3.66 string_escaping.libr.php

```
<?php

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```

```
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*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for HTML (surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\HTML
*/
namespace string_escaping\HTML {
  /**
  When you want that the string appears as is in HTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()


```

```
  /**
  When you want that the string appears as is in an HTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\, my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      ['"'],
      ['&quot;'],
      $s_string,
    );
  }//end escape_attribute()



  /**
  When you want that the string appears verbatim inside HTML <pre> tags.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\HTML
```

```
/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for XHTML (still surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\XHTML
*/
namespace string_escaping\XHTML {
  /**
  When you want that the string appears as is in XHTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()



  /**
  When you want that the string appears as is in an XHTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\, my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      [
        '&',
```

```
        '<',
        '"',
        "\n",
      ],
      [
        '&amp;',
        '&lt;',
        '&quot;',
        '&#10;',
      ],
      $s_string,
    );
  }//end escape_attribute()



  /**
  When you want that the string appears verbatim inside HTML <pre> tags.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\XHTML




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for JS (you're hopeless?)...

```

```
@package DevOrSysAdminScripts
@subpackage string_escaping\JS
*/
namespace string_escaping\JS {

  use function string_escaping\XHTML\escape_text as XHTML_escape_text;



  /**
  When you want that the string can be put in a JS string definition.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape(string $s_string) : string {
    return str_replace(
      [
        '\\',
        "\n",
        "'",
        '"',
      ],
      [
        '\\\\',
        '\n',
        '\\\'',
        '\\"',
      ],
      $s_string,
    );
  }//end escape()



  /**
  When you want that the string can be put in a JS string definition
  to be included inside an XHTML file <script> tag.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_for_string_definition_inside_XHTML(string $s_string)
  : string {
```

```
    return XHTML_escape_text(escape($s_string));
  }//end escape_for_string_definition_inside_XHTML()
}//end namespace string_escaping\JS




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for TeX.

@package DevOrSysAdminScripts
@subpackage string_escaping\TeX
*/
namespace string_escaping\TeX {
  /**
  When you want that the string can be written to a TeX file inside a
  part where TeX is in text mode.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    // https://tex.stackexchange.com/a/34586/270015
    return str_replace(
      [
        '\\',
        '&',
        '%',
        '$',
        '#',
        '_',
        '{',
        '}',
        '~',
        '^',
      ],
      [
        '\\textbackslash',
        '\\&',
        '\\%',
        '\\$',
        '\\#',
        '\\_',
```

```
        '\\{',
        '\\}',
        '\\textasciitilde',
        '\\textasciicircum',
      ],
      $s_string,
    );
  }//end escape_text()
}//end namespace string_escaping\TeX

/*
$ php
<?php
require_once('string_escaping.libr.php');
var_dump(string_escaping\HTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
  "abcd & < > \\ \n  \"'"
));
*/
?>
```

### 3.67 unicode.libr.php

```
<?php

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```
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This file was renamed from "unicode.php" to "unicode.libr.php".

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*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
Converts an int to the byte-string corresponding to the UTF-8 encoding
of the Unicode code point of same integer value.
decimal_code_point is a slight abuse of language to say that you write the
code point in decimal notation in PHP code and you get the UTF-8.
If you provide the integer of the code point without using decimal
notation, it will also work.

@param int $i_code_point_in_decimal_notation The code point.

@throws Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function decimal_code_point_to_UTF8(
  int $i_code_point_in_decimal_notation
) : string {
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
```

```
64    if($i_code_point_in_decimal_notation < 2048){  // 2**11
65      $i_first_byte_base_value = 192;
66      $i_last_byte_significant_bits = (
67        $i_code_point_in_decimal_notation % 64  // 2**6
68      );
69      $i_first_byte_significant_bits = intdiv(
70        (
71          $i_code_point_in_decimal_notation
72          - $i_last_byte_significant_bits
73        ),
74        64,  // 2**6
75      );
76      $s_result = chr(
77        $i_first_byte_base_value
78        + $i_first_byte_significant_bits
79      );
80      $s_result .= chr(
81        $i_continuation_base_value
82        + $i_last_byte_significant_bits
83      );
84      return $s_result;
85    }//end if($i_code_point_in_decimal_notation < 2048)
86    // 1110xxxx 10xxxxxx 10xxxxxx
87    if($i_code_point_in_decimal_notation < 65536){  // 2**16
88      $i_first_byte_base_value = 224;
89      $arr_arr_data_per_byte_reverse = [
90        ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
91        ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
92        ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],
93      ];
94      $i_rest = $i_code_point_in_decimal_notation;
95      $s_result = '';
96      foreach($arr_arr_data_per_byte_reverse as $arr_data){
97        // _Pragma unroll
98        // J'aurais pu "roller" les 4 cas des ifs.
99        // Mais pour les perfs, il faudrait tout unroller vu le nombre
100       // d'itérations. Et j'aimerais bien comparer du code unrollé
101       // aux 2 niveaux à la main avec celui unrollé par le compilateur
102       // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
103       // l'intérieur d'abord (pour ne pas refaire le même boulot)
104       // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
105       // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
106       $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
107       $i_rest = intdiv(
108         $i_rest - $i_significant_bits,
109         2**$arr_data['i_bits'],
```

```
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
    return $s_result;
  }//end if($i_code_point_in_decimal_notation < 65536)
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 2097152){  // 2**21
    $i_first_byte_base_value = 240;
    $arr_arr_data_per_byte_reverse = [
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = '';
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data['i_bits'],
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }
    return $s_result;
  }//end if($i_code_point_in_decimal_notation < 2097152)
  // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
```

```
    ." :)"
  );
  // phpcs:enable Squiz.Strings.DoubleQuoteUsage.NotRequired
}//end decimal_code_point_to_UTF8()



/**
Converts a string of hexadecimal digits to the byte-string corresponding
to the UTF-8 encoding of the Unicode code point of same integer value
that is denoted by the hexadecimal digits.

@param string $s_code_point_in_hexadecimal_notation The code point.

@throws Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function hexa_code_point_to_UTF8(
  string $s_code_point_in_hexadecimal_notation
) : string {
  $s_code_point_in_hexadecimal_notation = strtoupper(
    $s_code_point_in_hexadecimal_notation
  );
  $i_code_point_in_decimal_notation = 0;
  $i_ord_0 = ord('0');
  $i_ord_A = ord('A');
  $i_digit_offset = -$i_ord_0;
  $i_letter_offset = 10 - $i_ord_A;
  for(
    $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
    $i < $i_max;
    ++$i
  ){
    $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
    $i_code_point_in_decimal_notation *= 16;
    if(ctype_digit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
        ord($s_hexa_digit) + $i_digit_offset
      );
      continue;
    }
    if(ctype_xdigit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
        ord($s_hexa_digit) + $i_letter_offset
      );
```

```
202         continue;
203       }
204       throw new Exception('Not an hexadecimal digit.');
205     }
206     return decimal_code_point_to_UTF8(
207       $i_code_point_in_decimal_notation
208     );
209   }//end hexa_code_point_to_UTF8()
210
211
212
213   /**
214   This function throws an exception with extended debug informations
215   if the input string is not valid ASCII.
216   Otherwise, it returns true.
217
218   @param string $s_string The input string.
219
220   @throws Exception When the input string is not valid ASCII.
221
222   @return bool
223   */
224   function check_string_is_valid_ASCII(string $s_string) : bool {
225     $i_offset_in_octets_from_string_start = 0;
226     $i_current_line_number = 1;
227     $i_offset_in_octets_from_line_start = -1;
228     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
229       $i_current_octet = ord($s_string[$i]);
230       $i_offset_in_octets_from_string_start = $i;
231       ++$i_offset_in_octets_from_line_start;
232       if($i_current_octet > 127){
233         throw new Exception(
234           'Non-ASCII character found on line '
235           .$i_current_line_number
236           .'; the octet/character '
237           .($i_offset_in_octets_from_line_start + 1)
238           .' has value '
239           .$i_current_octet
240           .' which is not ASCII.'
241           .' (Sequential position without line splitting:'
242           .' This is at octet/character '
243           .($i_offset_in_octets_from_string_start + 1)
244           .'.)',
245         );
246       }
247       // if($s_string[$i] === "\n"){
```

```
248     if($i_current_octet === 10){
249       ++$i_current_line_number;
250       $i_offset_in_octets_from_line_start = -1;
251     }
252   }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
253   return true;
254 }//end check_string_is_valid_ASCII()
255
256
257
258 /**
259 This function throws an exception with extended debug informations
260 if the content of the file with given file_path is not valid ASCII.
261 Otherwise, it returns true.
262
263 @param string $s_file_path The input file_path.
264
265 @throws Exception When the input string is not valid ASCII.
266
267 @return bool
268 */
269 function check_file_is_valid_ASCII(string $s_file_path) : bool {
270   $s_string = file_get_contents($s_file_path);
271   if($s_string === false){
272     throw new Exception('File '.$s_file_path.' not found.');
273   }
274   return check_string_is_valid_ASCII($s_string);
275 }//end check_file_is_valid_ASCII()
276
277
278
279 /**
280 This function throws an exception with extended debug informations
281 if the input string is not valid UTF-8.
282 Otherwise, it returns true.
283
284 @param string $s_string The input string.
285
286 @throws Exception When the input string is not valid UTF-8.
287
288 @return bool
289 */
290 function check_string_is_valid_UTF8(string $s_string) : bool {
291   $i_continuation_octet_needed = 0;
292   $i_offset_in_octets_from_string_start = 0;
293   $i_offset_in_characters_from_string_start = -1;
```

```
294     $i_character_start_position_from_string_start = 0;
295     $i_current_line_number = 1;
296     $i_offset_in_octets_from_line_start = -1;
297     $i_offset_in_characters_from_line_start = -1;
298     $i_character_start_position_from_line_start = -1;
299     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
300       $i_current_octet = ord($s_string[$i]);
301       $i_offset_in_octets_from_string_start = $i;
302       ++$i_offset_in_octets_from_line_start;
303       if($i_continuation_octet_needed > 0){
304         if($i_current_octet < 128 || $i_current_octet >= 192){
305           throw new Exception(
306             'Non-UTF8 character found on line '
307             .$i_current_line_number
308             .'; the octet '
309             .($i_offset_in_octets_from_line_start + 1)
310             .', part of the character '
311             .($i_offset_in_characters_from_line_start + 1)
312             .', has value '
313             .$i_current_octet
314             .' which is not a continuation octet.'
315             .' This character starts at octet '
316             .($i_character_start_position_from_line_start + 1)
317             .' of the current line.'
318             .' (Sequential positions without line splitting:'
319             .' This is at character '
320             .($i_offset_in_characters_from_string_start + 1)
321             .' and octet '
322             .($i_offset_in_octets_from_string_start + 1)
323             .'.'
324             .' This character starts at octet '
325             .($i_character_start_position_from_string_start + 1)
326             .'.)',
327           );
328         }//end if($i_current_octet < 128 || $i_current_octet >= 192)
329         --$i_continuation_octet_needed;
330         continue;
331       }//end if($i_continuation_octet_needed > 0)
332
333       ++$i_offset_in_characters_from_string_start;
334       ++$i_offset_in_characters_from_line_start;
335       $i_character_start_position_from_string_start = (
336         $i_offset_in_octets_from_string_start
337       );
338       $i_character_start_position_from_line_start = (
339         $i_offset_in_octets_from_line_start
```

```
340       );
341       if($i_current_octet < 128){ // 0xxxxxxx ASCII
342         // if($s_string[$i] === "\n"){
343         if($i_current_octet === 10){
344           ++$i_current_line_number;
345           $i_offset_in_octets_from_line_start = -1;
346           $i_offset_in_characters_from_line_start = -1;
347         }
348         continue;
349       }
350       if($i_current_octet >= 128 && $i_current_octet < 192){
351         throw new Exception(
352           'Non-UTF8 character found on line '
353           .$i_current_line_number
354           .'; the octet '
355           .($i_offset_in_octets_from_line_start + 1)
356           .', part of the character '
357           .($i_offset_in_characters_from_line_start + 1)
358           .', has value '
359           .$i_current_octet
360           .' which is a continuation octet.'
361           .' This character starts at octet '
362           .($i_character_start_position_from_line_start + 1)
363           .' of the current line.'
364           .' (Sequential positions without line splitting:'
365           .' This is at character '
366           .($i_offset_in_characters_from_string_start + 1)
367           .' and octet '
368           .($i_offset_in_octets_from_string_start + 1)
369           .'.'
370           .' This character starts at octet '
371           .($i_character_start_position_from_string_start + 1)
372           .'.)',
373         );
374       }//end if($i_current_octet >= 128 && $i_current_octet < 192)
375       if($i_current_octet >= 192 && $i_current_octet < 224){
376         // 110xxxxx 10xxxxxx
377         $i_continuation_octet_needed = 1;
378         continue;
379       }
380       if($i_current_octet >= 224 && $i_current_octet < 240){
381         // 1110xxxx 10xxxxxx 10xxxxxx
382         $i_continuation_octet_needed = 2;
383         continue;
384       }
385       if($i_current_octet >= 240 && $i_current_octet < 248){
```

```
      // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
      $i_continuation_octet_needed = 3;
      continue;
    }
    throw new Exception(
      'Non-UTF8 character found on line '
      .$i_current_line_number
      .'; the octet '
      .($i_offset_in_octets_from_line_start + 1)
      .', part of the character '
      .($i_offset_in_characters_from_line_start + 1)
      .', has value '
      .$i_current_octet
      .' which is invalid.'
      .' This character starts at octet '
      .($i_character_start_position_from_line_start + 1)
      .' of the current line.'
      .' (Sequential positions without line splitting:'
      .' This is at character '
      .($i_offset_in_characters_from_string_start + 1)
      .' and octet '
      .($i_offset_in_octets_from_string_start + 1)
      .'.'
      .' This character starts at octet '
      .($i_character_start_position_from_string_start + 1)
      .'.)',
    );
  }//end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
  return true;
}//end check_string_is_valid_UTF8()



/**
This function throws an exception with extended debug informations
if the content of the file with given file_path is not valid ASCII.
Otherwise, it returns true.

@param string $s_file_path The input file_path.

@throws Exception When the input string is not valid UTF-8.

@return bool
*/
function check_file_is_valid_UTF8(string $s_file_path) : bool {
  $s_string = file_get_contents($s_file_path);
```

```
  if($s_string === false){
    throw new Exception('File '.$s_file_path.' not found.');
  }
  return check_string_is_valid_UTF8($s_string);
}//end check_file_is_valid_UTF8()



/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/
```

```
?>
```

### 3.68 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --fixed-point-build
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
```

```
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "."\
    "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"\
    "${LFBFL_check_download}"
}

upgrade_build_and_checks "$@"
```

### 3.69 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

```

```
27 wget_sha512(){
28   # $1=file_path
29   # $2=download_URL
30   # $3=correct_sha512
31   # Options:
32   #   --check-download (=keep to keep the downloaded file)
33   #   --verbose
34   local LFBFL_verbose=""
35   if [[ "$*" == *--verbose* ]]; then
36     printf "%s wget_sha512 %s\n" "$0" "$*"
37     LFBFL_verbose="--verbose"
38   fi
39   readonly LFBFL_verbose
40   local LFBFL_check_download=""
41   if [[ "$*" == *--check-download=keep* ]]; then
42     LFBFL_check_download="--check-download=keep"
43   elif [[ "$*" == *--check-download* ]]; then
44     LFBFL_check_download="--check-download"
45   fi
46   readonly LFBFL_check_download
47   declare -i LFBFL_i_error=0
48
49   if [[ ! -f "$1" ]]; then
50     wget ${LFBFL_verbose:+"${LFBFL_verbose}"} -O "$1" "$2"
51   fi
52   declare -r LFBFL_present_sha512=$(
53     sha512sum "$1"\
54     | cut --fields=1 --delimiter=' '
55   )
56   if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
57     printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
58       "$1" "$3" "${LFBFL_present_sha512}"
59     LFBFL_i_error=1
60   fi
61   if [[ -n "${LFBFL_check_download}" ]]; then
62     wget ${LFBFL_verbose:+"${LFBFL_verbose}"} -O "$1.wget_sha512.temp" "$2"
63     declare -r LFBFL_present_sha512_2=$(
64       sha512sum "$1.wget_sha512.temp"\
65       | cut --fields=1 --delimiter=' '
66     )
67     if [[ "${LFBFL_check_download}" != "--check-download=keep" ]]; then
68       rm "$1.wget_sha512.temp"
69     fi
70     if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
71       printf\
72         "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
```

```
        "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
      LFBFL_i_error=1
    fi
  fi
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !